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Stress analysis of dual-clutch transmission components through lightweighting using a finite-element analysis approach

BEng PROJECT FINAL DISSERTATION 2024-25

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Summary

As automotive manufacturers strive to reduce vehicle weight and improve drivetrain efficiency, lightweighting of automotive components has become a critical area of research. One such component is the input shaft of a dual-clutch transmission (DCT), which experiences significant torsional loads during operation. This Project explores the potential of high-strength steel alloys to reduce weight and enhance the performance of DCT input shafts through material substitution.

The central aim of this work was to conduct a comparative analysis of different materials to determine their suitability for DCT input shafts based on mechanical strength, weight, torsional stiffness, and cost. The study focused on evaluating the feasibility of material substitution using a finite-element approach. Finite element analysis (FEA) was conducted in Abaqus CAE on the inner and outer input shafts under a torque load of 420 Nm. Different metal alloys were evaluated, and the accuracy of the simulations was ensured by validating them against classical torsion theory.

Key results showed that although variations in stress levels were negligible across all materials, the factor of safety (FoS) varied widely which was dependent on the material's yield strength. Some materials showed FoS values more than double that of the control baseline material. Most materials exhibited minimal differences in torsional stiffness, confirming that stiffness is not a limiting constraint.

The study concludes that strategic material substitution alone can significantly improve strength and reduce the cost of DCT shafts. A Materials Performance Index (MPI) was used to rank materials based on performance trade-offs, aiding decision-making. The methodology developed here provides a framework for future component optimisation.

Recommendations for future work include evaluating fatigue life under cyclic loading, exploring geometry-based lightweighting, and conducting thermal loading analysis. Additionally, sustainability considerations such as embodied energy and recyclability could further guide material choices for next-generation automotive transmissions.

Contents

NOMENCLATURE

Symbol	Quantity	Unit
C	Cost	£
C_v	Cost Per Volume	$£/m^3$
D	Diameter	m
E	Young's Modulus	GPa
G	Shear Modulus	GPa
J	Polar Moment of Area	m^4
k	Stiffness	N/m
L	Length	m
m	Mass	kg
N_x	Normalised Value	
r	Radius	m
T	Torque	Nm
V	Volume	m^3
W_x	Weightings	
θ	Twist Angle	rad
ρ	Density	kg/m^3
τ	Torsional Stress	Pa
σ_f	Fatigue Strength at 10^7Cycles	MPa
σ_{Max}	Maximum von Mises Stress	MPa
σ_y	Yield Stress	MPa
σ_1	Maximum Principal Stress	MPa
σ_3	Minimum Principal Stress	MPa

Subscripts & Superscripts

FEA	FEA Derived Values	
i	Inner	
o	Outer	
$THEO$	Theoretical Derived Values	

ACRONYMS

Acronym	Meaning
AHSS	Advanced High Strength Steel
AISI	American Iron and Steel Institute
AT	Automatic Transmission
BMW	Multinational manufacturer of vehicles and motorcycles
CAD	Computer-Aided Design
CAE	Computed-Aided Engineering
CES	Cambridge Engineering Selector
DCT	Dual-Clutch Transmission
DSG	Direct-Shift Gearbox, is VW's trademarked DCT
FEA	Finite-Element Analysis
FoS	Factor of Safety
HSS	High Strength Steel
MPI	Materials Performance Index
MT	Manual Transmission
.STEP	A widely used data exchange file used in CAD
TCU	Transmission Control Unit
UHSS	Ultra High Strength Steel
VW	Volkswagen AG is a conglomerate car manufacturer
x-Speed Gearbox	Gearbox with "x" number of forward gears
ZF	ZF Group is an automotive technology supplier

1. BACKGROUND AND CONTEXT

The automotive industry faces stricter emissions standards, with the design and manufacturing of vehicles influenced by progressively firmer regulations. Regulatory frameworks such as the European Union and the USA's Corporate Average Fuel Economy standards have introduced targets to reduce fuel consumption by 2025 and 2030 [1]. The push to dramatically improve transport efficiency has led the EU to push for a 100% reduction in emissions by 2035 [2]. Emissions need a 90% reduction to meet the EU's 2050 climate neutrality target [3]. Additionally, consumer demand for superior speed, handling, and acceleration has also led to ever-evolving technology that satisfies the personal feeling experienced when driving high-performance cars. Balancing environmental goals and performance expectations highlights the need for innovative solutions. One effective way is the lightweighting of motor vehicles, as demonstrated in the evolution of automotive transmissions, where advancements have generated fuel-efficient and high-performance solutions [4].

The manual transmission's (MT) simplicity and control are key factors that contribute to its continued widespread use in modern vehicles. Using a driver-controlled clutch pedal, the driver can manually modulate the power supply from the engine to the transmission, where different gears can be selected to optimise power delivery to the wheels [5]. The MT's low-cost, straightforward design appeals to many drivers and car enthusiasts [5]. However, the MT's lack of seamless power delivery led to the more convenient hydraulic automatic transmission (AT), which involves automated gear changes. Whilst the AT did not improve efficiency compared to an MT, the AT provided automated and seamless gear changes. The lack of efficiency was due to using a fluid-coupled torque converter between the engine and gearbox. A torque converter uses a pump, turbine, and stator to multiply torque from the engine to the gearbox and through planetary gear sets, different gears can be engaged to provide continuous power delivery, as seen in Figure 1:

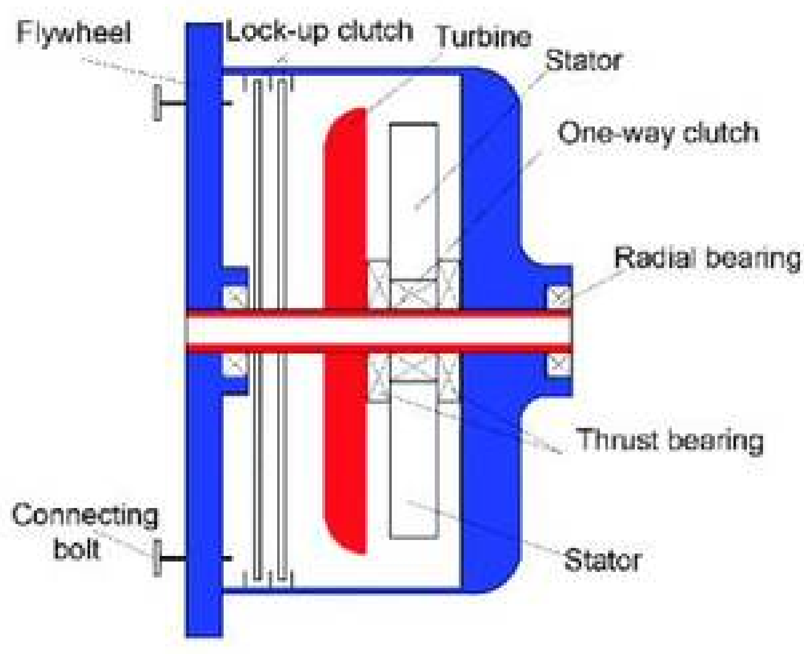

Figure 1: Schematic representation of Hydraulic Torque Converter Automatic Transmission illustrating the layout of key components [6].

While using a torque converter improved the driving quality, the energy losses caused by slippage at low speeds drastically increased fuel consumption [7]. In response to these issues, the dual-clutch transmission (DCT) was developed, which was an advanced improvement of the AT. A DCT utilises two clutches where one clutch engages the odd-numbered gears (1st, 3rd, 5th Gear or higher) and the other clutch engages the even-numbered gears (2nd, 4th, 6th Gear or higher), as seen in Figures 2 and 3:

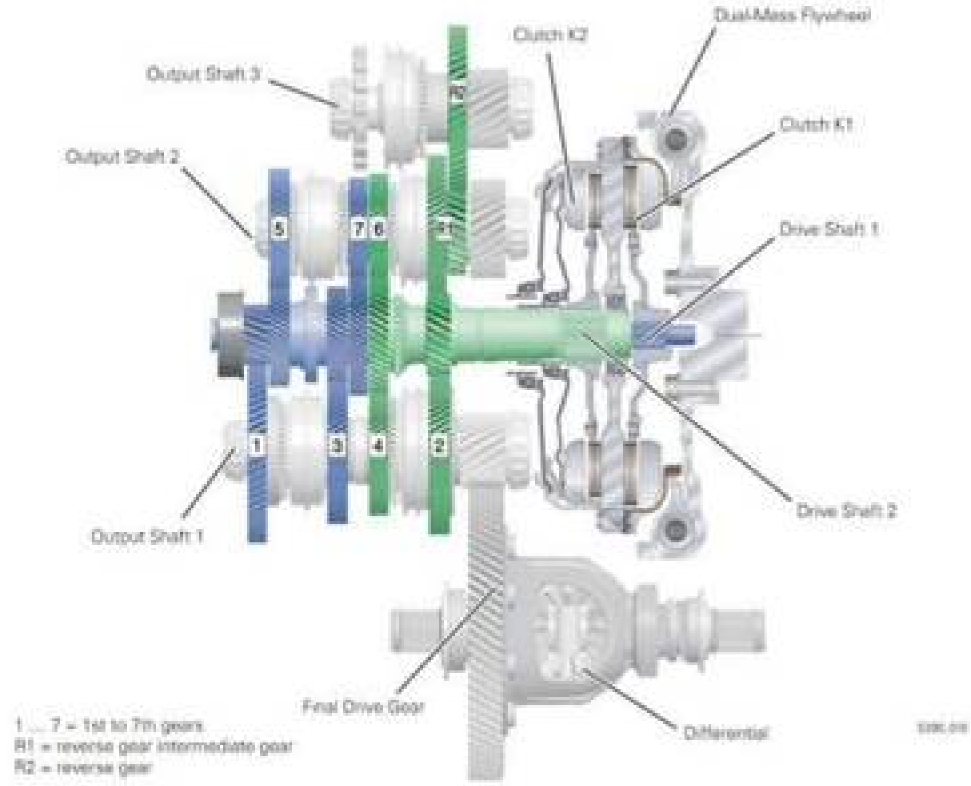

Figure 2: Schematic representation of VW's 7 Speed DSG, illustrating the layout of key components [8].

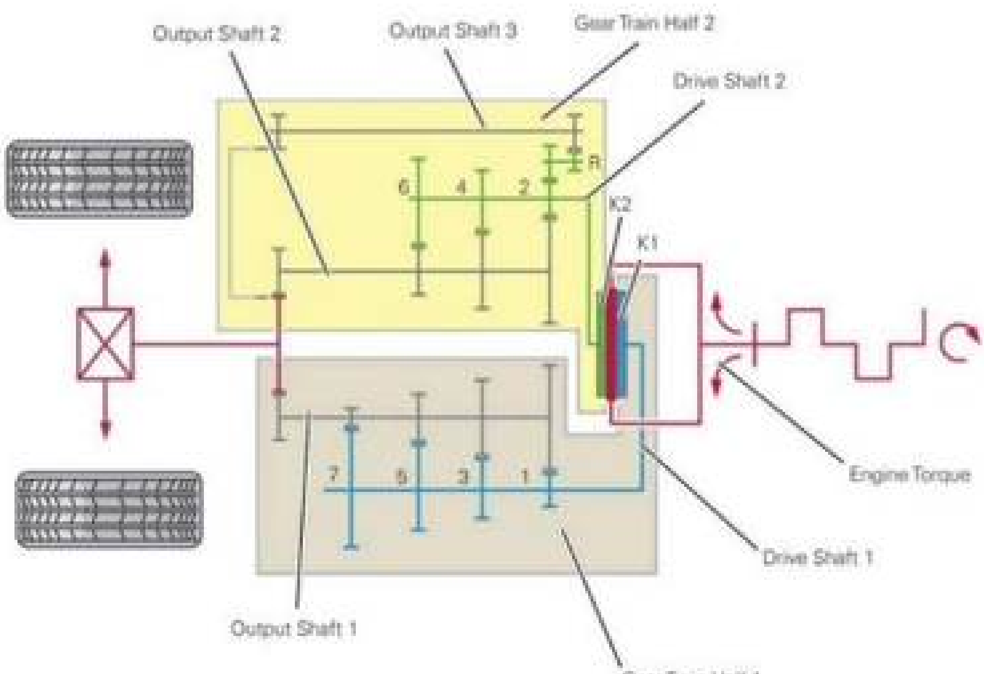

Figure 3: Diagram of VW's 7 Speed DSG showing the flow of engine torque through the system [8].

Each clutch connects to independent and concentric input shafts, where the outer shaft is hollow and encircles the inner shaft. This setup allows one clutch to transmit torque to the wheels through one shaft whilst the other clutch and shaft have another gear preselected. When the Transmission Control Unit (TCU) determines the optimum shifting time, dependent on engine load, speed, driver input and throttle position, it disengages one clutch, whilst simultaneously engaging the other clutch. Using a TCU introduced an innovative solution with advanced algorithms to optimise clutch and gear engagement, delivering unprecedented intelligence and precision not used in previous automotive transmissions, which allowed millisecond gear shifting and seamless power delivery. The mechanical linkage from the engine flywheel through the gearbox allows for minimal energy losses common in automatic fluid-coupled torque converters. DCTs give the driver the best of both worlds.

Porsche pioneered the DCT in their 1970s racing cars, a groundbreaking innovation despite challenges arising from the limited electronic and hydraulic technology during that time [9]. Over the years, development in technology led to the first mass-produced DCT for the Mark IV Volkswagen Golf R32 in 2003. It brought about quicker acceleration due to faster shifting than its manual counterpart and reduced fuel consumption [10]. VW claimed that CO_2 emissions reduced from 149 g/km to 139 g/km [11]. Developments have reduced shift times in a DCT, such as 0.2 seconds to change gears [12] in a Nissan GTR's DCT, which weighed 117kg [13]. The short time taken to change gears can explain why the number of DCT-equipped cars is expected to grow at a compound annual growth rate of 4.9% between 2021 and 2027 [14]. However, despite the benefits, they are heavier and more expensive than other transmissions, restricting their application in the market. BMW expressed this by dropping its ZF-manufactured DCT in its M Cars, claiming improved fuel efficiency and drivability in current-generation torque converters [15]. Lightweighting can seek to push these limits by addressing the DCT's drawbacks.

This Project focuses on analysing the torsional performance of input shafts within a DCT to address structural challenges present in current designs. Finite-Element Analysis (FEA) techniques will simulate real-world torsional loading conditions, allowing evaluation of the proposed shaft geometries under operational stresses. The primary emphasis of this study is on lightweighting strategies that can integrate into existing DCT systems without necessitating complete redesigns. In particular, the Project will investigate whether high-strength alloys can reduce weight while maintaining or improving structural integrity, offering a practical solution for enhancing current transmission systems.

1.1. Aim and Objectives

The Project aims to study the stress distribution in a DCT's input shafts using an FEA approach. Different grades of advanced high-strength alloys will be evaluated to enhance the DCT's efficiency and performance.

Objectives

- Develop a SolidWorks model of the proposed DCT input shafts for FEA simulations to predict real-world applications and analyse the stress distributions.
- Simulate torsional loading conditions in Abaqus CAE to evaluate shear stress distributions and the shaft's structural performance.
- Validate FEA models and simulations against theoretical data to ensure the accuracy of results.
- Analyse and evaluate different high-grade alloys viable for use in the DCT input shafts, which aim to reduce weight while enhancing the DCT's performance.
- Compare the proposed design with current designs to quantify any improvements in weight or cost.

2. LITERATURE REVIEW

2.1. Dual Clutch Transmission Input Shafts

The input shafts of a DCT play a vital role in the operation of the transmission system. In a DCT, the input shafts transmit rotational power from the engine to the gears, where typically, a DCT utilises two concentric shafts: an inner shaft and an outer shaft. The shafts experience high torsional loads due to the torque generated by the engine [16]. These also include variations of torque applied to the input shafts in different scenarios, such as rapid acceleration or downshifting. Hard acceleration from rest can mean applied torque surges above the rated peak. Poorly designed input shafts can plastically deform in these scenarios, leading to eventual failure. Evaluating the shafts' design must consider these factors [16]. The shafts also experience cyclic loading due to the loading and unloading of torque, which introduces the risk of fatigue failure. The heat generated in the gearbox due to friction can induce thermal stresses in components. Peak engine torque applied at the input shafts can range from 200Nm in passenger cars to 1000Nm and above in hypercars such as the Pagani Huayra [17].

DCTs are expected to last well over 200,000 miles, so the input shafts must be engineered to withstand these high forces for a long time. The development of stress concentrations in areas such as splines, keyways, and diameter transitions further complicates the challenge of designing an input shaft. Splines, which facilitate torque transfer between the shaft and the gears or clutch whilst allowing axial movement, often experience high shear stresses. Poorly designed splines or sharp transitions between shaft sections can result in stress concentrations, increasing the likelihood of failure. To mitigate these failures, design elements such as fillets and optimised spline profiles are incorporated to distribute stress more evenly [18]. While extensive research has been conducted into optimising gears and shafts in other applications, there is limited open-source research into lightweighting concentric input shafts of a DCT. Figures 4, 5 and 6 show the input shaft assembly, highlighting the primary stress concentration and how the shafts are aligned:

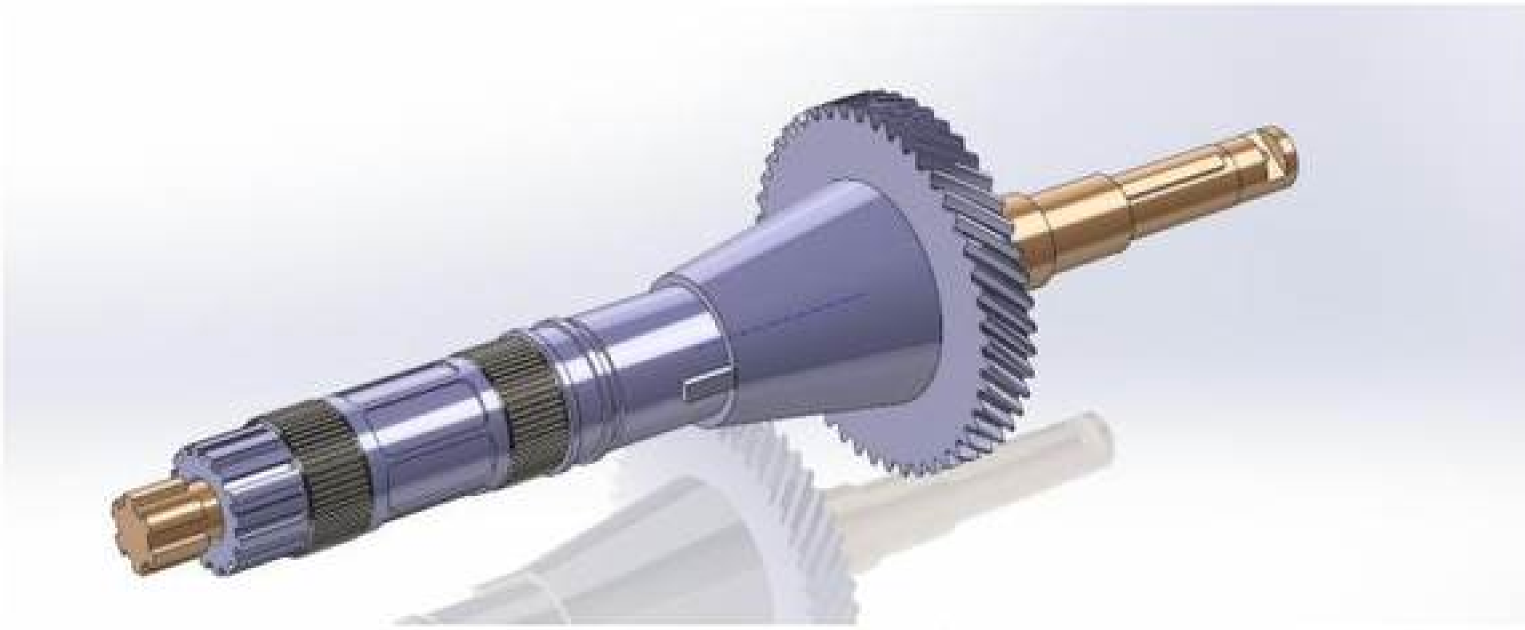

Figure 4: CAD Model of a DCT's Drive Shaft Assembly based on an open-source CAD model [19].

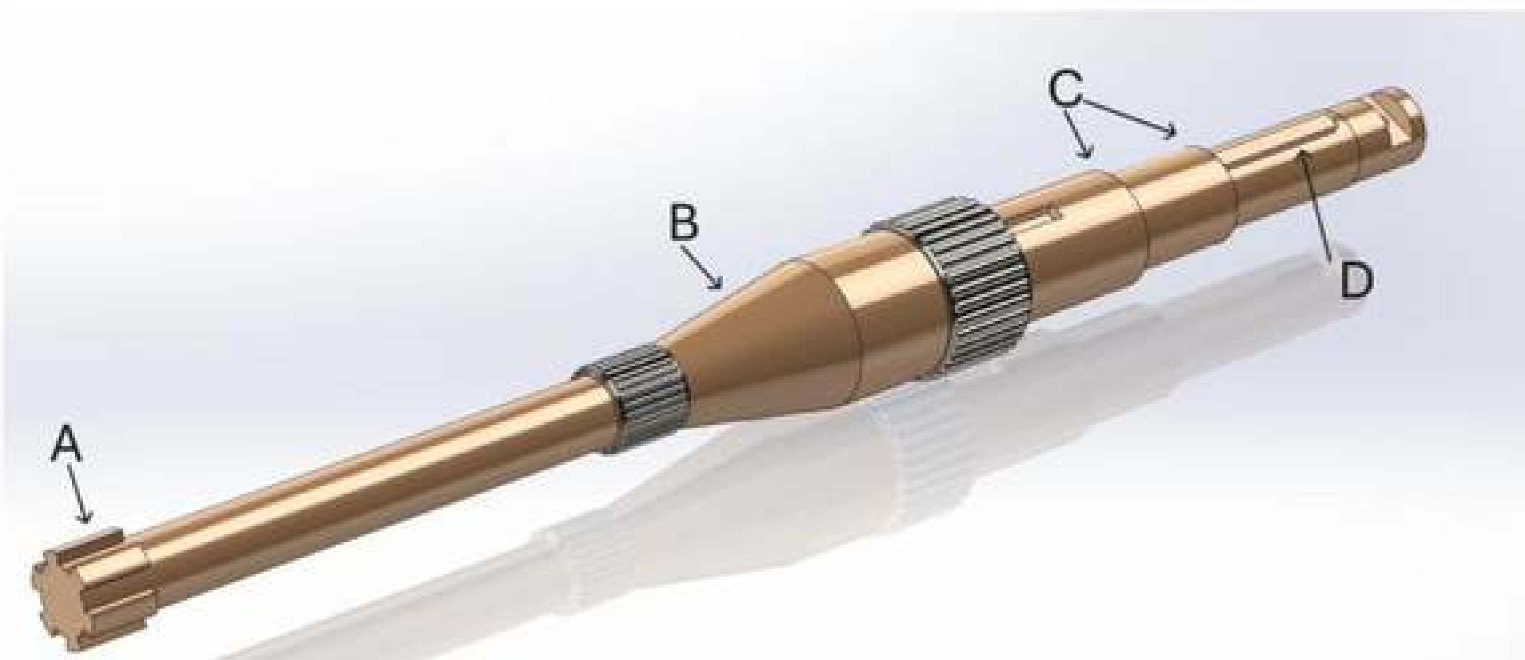

Figure 5: CAD Model of Inner Drive Shaft based on an open-source CAD model [19].

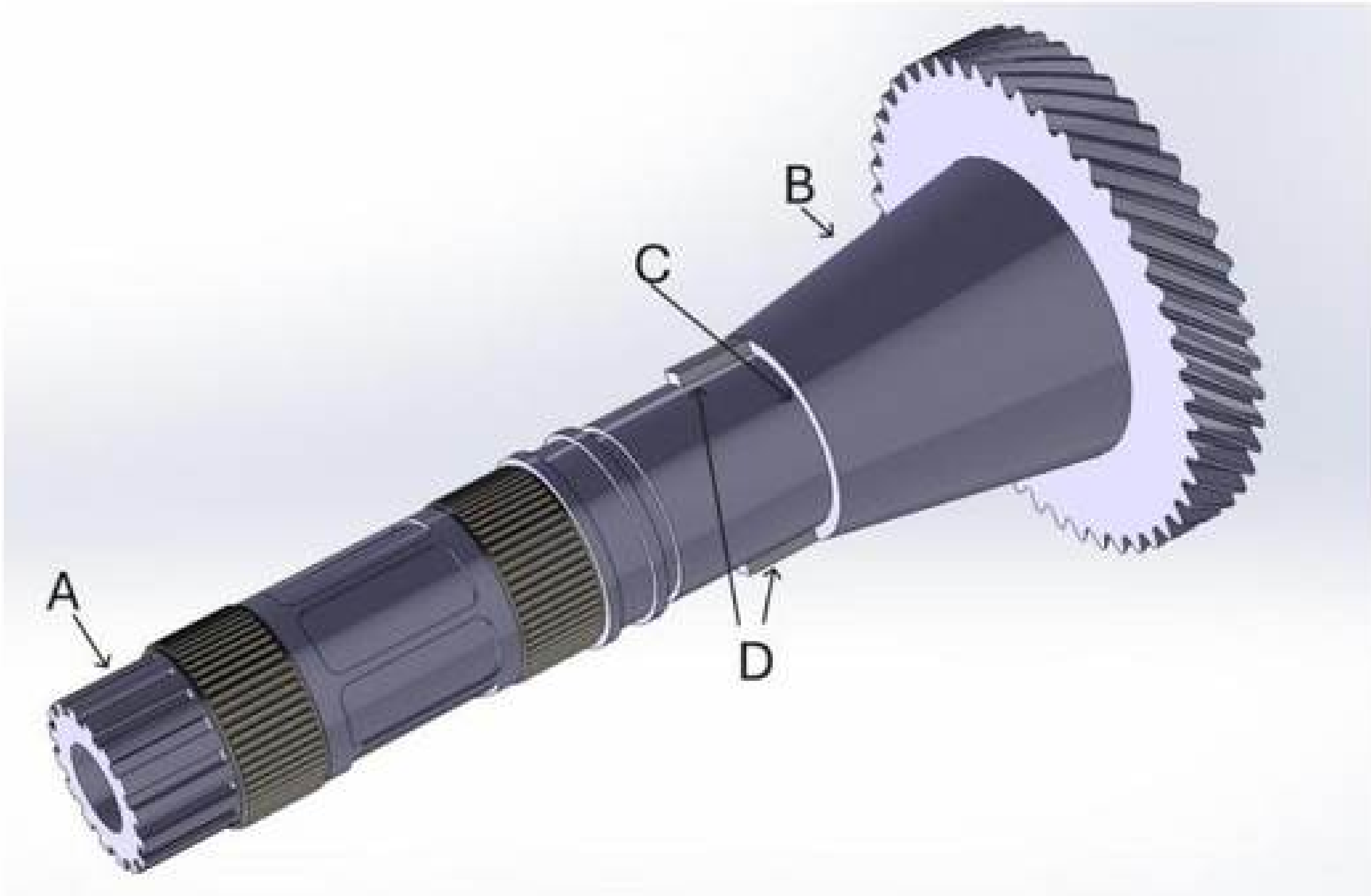

Figure 6: CAD Model of Outer Drive Shaft based on an open-source CAD model [19].

In Figures 5 and 6, Point A shows the splined region of the shafts where the shaft connects to the clutch pack. Point B shows a gradual diameter transition where stress concentrations are expected to be low. Point C shows the sharp diameter transitions between regions, which are also visible between other regions. Finally, Point D shows the location of keyways on the shaft.

2.2. Materials

Optimising input shafts for a DCT requires grades with high shear strength and a high strength-to-weight ratio. They must also have a high fatigue life to be as durable as current designs. Selecting grades for DCT input shafts must account for resistance to cyclic and impulsive loads, as these factors significantly impact fatigue life. Al-Shammaa and Kadhim [20] demonstrated that stress distributions at spline roots under cyclic loading conditions can lead to premature failure if materials lack adequate fatigue strength. The proposed materials must be able to handle torques generated by the engine whilst being lightweight and sustainable. The current range of densities of automotive grade steels is between 7000 and 8000 kg/m^3, and the prospective material must retain its strength whilst having a low density. The common materials used in current designs are steel alloys due to their high tensile yield strength, and low cost [21]. High Strength Steels (HSS) and Ultra High Strength Steels (UHSS), with yield strengths ranging from 210 MPa to over 550 MPa, offer significant advantages in strength-to-weight ratio for automotive applications, including DCT input shafts.

Steel can also be strengthened through various mechanisms that prevent dislocation motion, enhancing mechanical performance. These methods include solid solution strengthening, where alloying elements create localised strain fields and grain refinement, which increases strength by introducing more grain boundaries. Another method is work hardening, which entangles dislocations through plastic deformation. Precipitation hardening introduces fine particles like carbides or nitrides that hinder dislocation movement, while transformation strengthening involves creating complex microstructures such as martensite or bainite during controlled cooling processes. These techniques are fundamental to developing advanced high-strength steels, such as dual-phase and transformation-induced plasticity steels. In the context of transmission drive shafts, these strengthening mechanisms significantly enhance yield strength, fatigue life, and torsional stiffness, which are key properties that resist high cyclic torque loads. Manufacturers can reduce shaft diameter and weight by using Advanced High Strength Steels (AHSS) with improved microstructures without sacrificing strength or durability. This improves performance in modern high-load transmission systems [22].

However, AHSS may present challenges in formability for complex geometries like splines and transitions [23]. Aluminium and magnesium alloys are also standard in applications where lightweight components are required, but these alloys fail to meet the strength and stiffness requirements of specific torque applications. Keyed shaft-hub connections provide an insight into the shafts' material strength limitations under torsional loads. Low-strength materials typically fail due to excessive surface pressure at the keyway, whereas high-strength materials often fail through shaft cracking before critical keyway expansion [24]. This highlights that while higher-strength materials improve resistance to deformation, they do not proportionally enhance durability in cyclic loading scenarios, with failure occurring at comparable permissible torque levels [24].

Understanding these failure mechanisms is critical for selecting materials and designing DCT input shafts, where fatigue strength and stress distribution must be optimised to ensure long-term reliability under high loads.

2.3. Finite-Element Analysis

FEA is a computational tool used to predict different structural loading scenarios by solving complex differential equations. The number of physical prototypes manufactured can be reduced efficiently by optimising designs in software using FEA. This speeds up the development of components by visualising how they would perform in real-world environments without physically needing to test them. Components are divided into discrete elements, meshed and instanced into an assembly. These assemblies then have boundary conditions applied to resemble operational constraints. Stresses and strains can then be visualised. In the context of DCT input shafts, different material properties can be input into the FEA software, and torsional stresses can be predicted by specifying input parameters, such as torque and boundary conditions, without ordering material or manufacturing prototypes. This provides an efficient way of evaluating varied materials to understand their suitability. This Project will make use of Abaqus CAE. Most open-source research utilises a consistent methodology, especially regarding boundary conditions and meshing [25]. Whilst no information is available specific to the input shafts of a DCT, the torsional theory in FEA for any generic shaft can still be applied to this Project. Jayanaidu et al. [26] demonstrated the advantages of titanium alloys over conventional steel in drive shafts, showing reduced deformation and weight while maintaining high strength. Their FEA-based approach and modal analysis provide a robust framework for evaluating lightweight materials in automotive shafts. The FEA simulations must be validated, as validation against classical torsional theory equations is standard practice to ensure simulation reliability.

3. METHODOLOGY

3.1. Computer-Aided Design Modelling of Input Shaft Assembly

Computer-Aided Design (CAD) Models of the inner and outer input shafts were modelled in SolidWorks using an open-source CAD model as a reference [19]. Due to the proprietary nature of automotive systems and components, manufacturers do not typically release specifications and component dimensions as open-source. Current open-source knowledge relies on independent researchers or enthusiasts who purchase production components, disassemble them, and publish information on the dimensions of the components or as open-source CAD models. Figures 7 and 8 below show the CAD models of the inner and outer shafts, where each shaft has been sectioned into regions of different diameters:

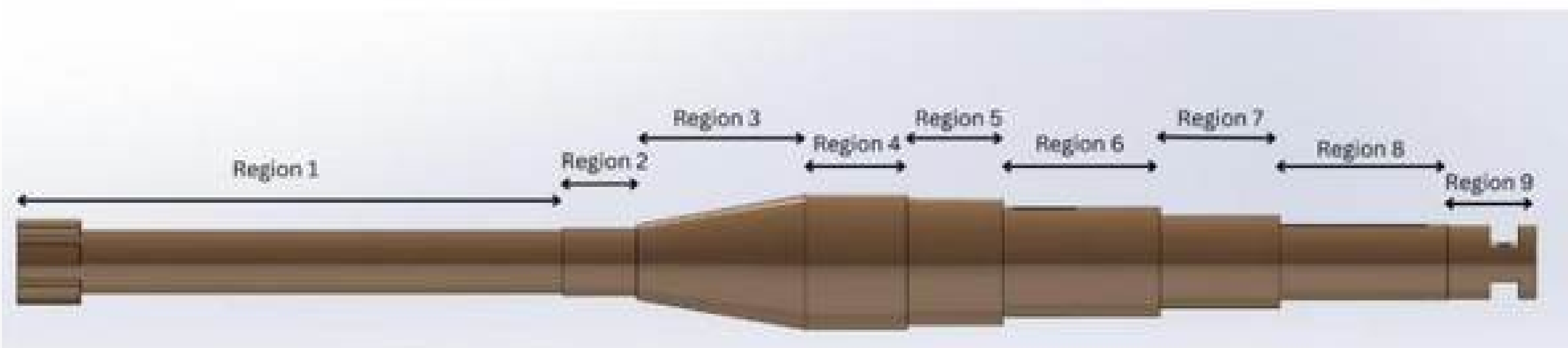

Figure 7: Inner Input Shaft showing how the shaft was sectioned into 9 regions.

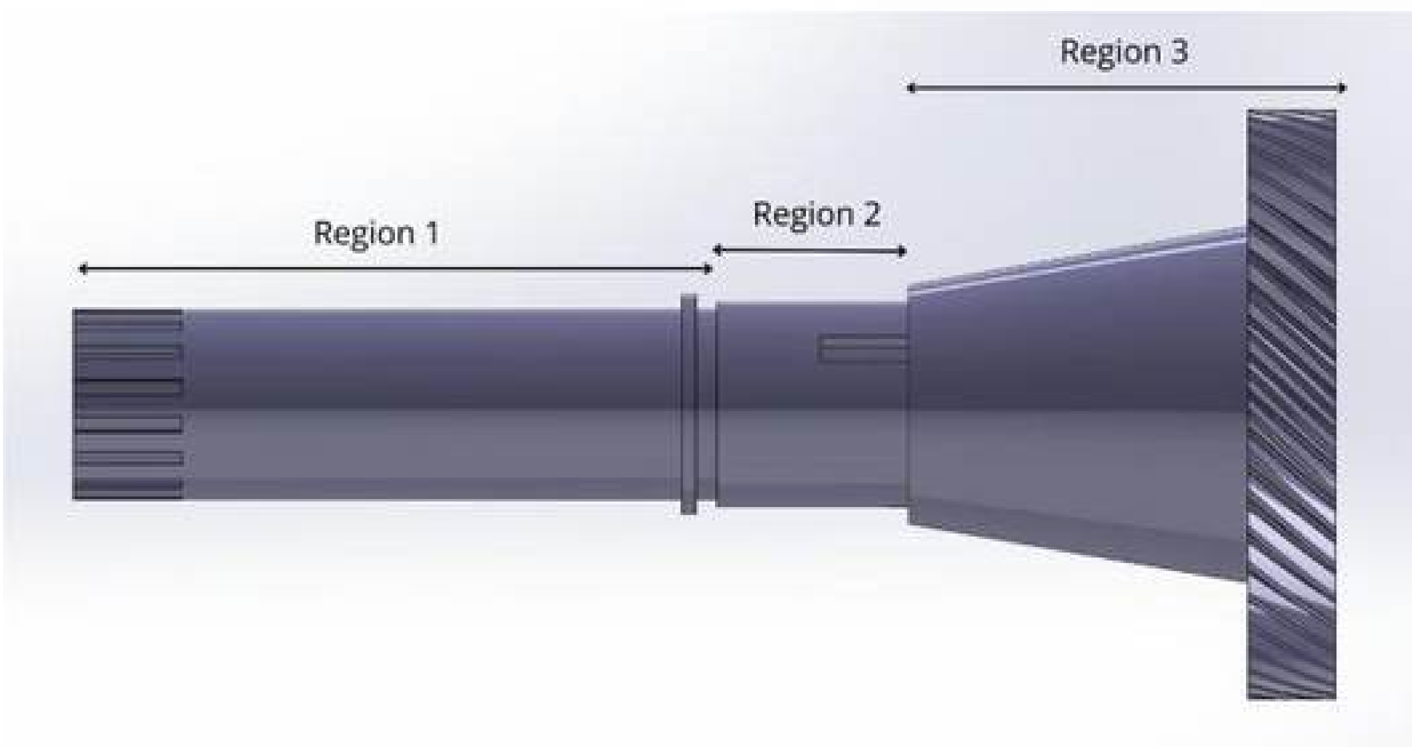

Figure 8: Outer Input Shaft showing how the shaft was sectioned into 3 regions.

Tables 1 and 2 below detail the critical dimensions of both the inner and outer shaft:

Table 1: Critical dimensions of the inner input shaft, listed by region for length and diameter.

Inner Shaft

Feature	Region	Length (mm)	Diameter (mm)	Notes
Spline	Region 1	260	30	Spline not considered
-	Region 2	35.5	32	
Taper	Region 3	80.4	35.6 → 64.0	Smaller to larger diameter of the taper
-	Region 4	48.5	64	
-	Region 5	45.6	60	
Keyway	Region 6	75	52	Keyway not considered
-	Region 7	57.2	44	
Keyway	Region 8	80	36	Keyway not considered
Slot	Region 9	42	35	Slot not considered

Table 2: Critical dimensions of the outer input shaft, listed by region for length and diameter.

Outer Shaft

Feature	Region	Length (mm)	Diameter (mm)	Position	Notes
Spline	Region 1	220	65	D_o	Spline not considered
			34	D_i	
Spline	Region 2	65.5	70	D_o	
			34	D_i	
Taper	Region 3	142.3	81.4 → 122	D_o	Smaller to larger diameter of the taper
			34	D_i	

The dimensions and relations between the transmission components were understood using an open-source CAD model. The sectioning of the shafts allows for defined reference points of different regions for subsequent theoretical calculations of torsional shear stress and angle of twist. The inner input shaft was modelled with key features, including many sharp diameter changes, splines, and keyways. These features are considered critical areas of interest due to the high-stress concentrations and will be key for FEA simulations. Region 1, referencing the shaft's minimum diameter of 30 mm, is important, as regions of smaller diameter are expected to experience higher torsional stresses under applied torque. The outer input shaft was also modelled with key features such as a hollow bore with an internal diameter of 34mm, splines, and keyways.

Once CAD modelling was completed, the models were exported independently as .STEP files, which were preferred over alternative options due to their ability to store precise geometric and parametric data. Furthermore, .STEP files also store solid modelling information, which makes them better for CAE applications.

3.2. Materials Evaluation

Different steel grades were evaluated using a Materials Performance Index (MPI). An MPI provides a quantifiable ranking based on various material properties, which allows for simpler material selection. The higher a material's MPI, the better suited it is for the application and objectives.

The following material properties will be used to evaluate different grades:

- Yield Strength: Resistance to permanent deformation under torsional loads.
- Density: Important in lightweighting.
- Shear Modulus: Measure of the torsional rigidity of a material.
- Young's Modulus: Measure of elasticity and indicates axial stiffness.
- Cost: Assess economic viability.

The MPI formula is shown below in Equation 1:

$$MPI = W_1\sigma_y + W_2\frac{1}{\rho} + W_3G + W_4\frac{1}{C} + W_5\sigma_f \qquad \text{Equation 1}$$

where W_1 (0.15), W_2 (0.3), W_3 (0.2), W_4 (0.2) and W_5 (0.15) are weights to bias parameters depending on each criterion's relative importance.

- σ_y is the Yield Stress of the material,
- ρ is the Density of the material,
- G is the Shear Modulus of the materials,
- C is the Cost per Weight of the material,
- σ_f is the Average Fatigue Strength for 10^7 cycles.

Each material property is normalised through Equation 2 below:

$$Normalised\ Property\ Value = \frac{Property\ Value\ For\ Specific\ Material}{Maximum\ Property\ Value\ Across\ All\ Materials} \quad \text{Equation 2}$$

The normalised properties allow the property values to be on the same scale and dimensionless.

Table 3 presents the shortlisted materials identified as the most suitable for the application, along with any treatments and processing methods applied to enhance their performance:

Table 3: Selected candidate materials and their associated heat treatments or processing conditions.

Steel Grade	Treatment
Carbon Steel AISI 1030	Water Quenched & Tempered at 425°
Carbon Steel AISI 1045	Cold Drawn / As Rolled
Carbon Steel AISI 1080	As Rolled
Carbon Steel AISI 1095	Oil Quenched & Tempered at 650°
Complex Phase Steel, YS800	Cold Rolled
High Strength Low Alloy Steel YS550	Hot Rolled
Low Alloy Steel 50B46	Oil Quenched & Tempered at 315°
Low Alloy Steel AISI 4140	Oil Quenched & Tempered at 315°
Low Alloy Steel AISI 8740	Oil Quenched & Tempered at 425°
Maraging Steel 300	Maraged at 482°
Stainless Steel , Martensitic ASTM CA15	Cast Tempered at 790°
Stainless Steel 17-4PH, H1100	As Rolled
Stainless Steel, Martensitic, AISI 410	Intermediate Temper

Material property data was sourced from CES EduPack 2019, a comprehensive materials database widely used in engineering design and selection. However, most material data was averaged because CES EduPack 2019 collates data from multiple sources and provides a range for each material property due to small disparities in data. For example, the cost per kilogram for a material is an estimate as it can fluctuate depending on market conditions and sourcing region. All material properties were averaged from the range provided by CES EduPack. Therefore, these properties may not fully capture the exact performance of a material under all real-world conditions. These limitations are acknowledged, and all results should be interpreted considering these assumptions. Table 4 below details the critical material properties to be input into the MPI equation:

Table 4: Critical material properties for different materials [27-32].

Material	σ_y (MPa)	E (GPa)	G (GPa)	$Poisson's$ $Ratio$	ρ (kg/m^3)	$Cost$ $(£/kg)$	σ_f (MPa)
AISI 1030	580	212	82.5	0.29	7850	4.40	453.00
AISI 1045	530	205	80	0.29	7850	0.59	640.00
AISI 1080	587.5	207.5	80.5	0.29	7850	0.58	344.00
AISI 1095	552.5	207.5	80.5	0.29	7850	0.58	304.50
YS800	800	210.5	80.85	0.3	7850	0.58	403.50
YS550	600	210.25	81.1	0.3	7850	0.57	396.50
50B46	1620	206.5	80	0.29	7850	0.68	550.00
AISI 4140	1432.5	212	82.5	0.29	7850	0.98	447.48
AISI 8740	1357.5	206.5	80	0.29	7850	0.63	581.00
Maraging Steel 300	1901.06	187.69	71.79	0.32	7916.45	0.99	330.00
ASTM CA15	515	200	78	0.28	7610	27.99	856.50
17-4PH, H1100	870.5	202	90.25	0.28	7860	0.57	297.50
AISI 410	775.5	200	78	0.28	7750	1.12	313.00

The normalised values to be entered into the MPI are detailed below in Table 5:

Table 5: Normalised material properties for the MPI.

Material	$N\ \sigma_y$	$N\ \rho$	$N\ G$	$N\ Cost$	$N\ \sigma_f$	MPI
AISI 1030	0.31	0.97	0.91	0.99	0.40	0.78
AISI 1045	0.28	0.97	0.89	0.99	0.36	0.76
AISI 1080	0.31	0.97	0.89	0.99	0.47	0.78
AISI 1095	0.29	0.97	0.89	1.00	0.46	0.78
YS800	0.42	0.97	0.90	0.51	0.37	0.69
YS550	0.32	0.97	0.90	1.00	0.35	0.77
50B46	0.85	0.97	0.89	0.97	0.75	0.90

AISI 4140	0.75	0.97	0.91	0.91	0.68	0.87
AISI 8740	0.71	0.97	0.89	0.84	0.64	0.55
Maraging Steel 300	1.00	0.96	0.80	0.02	1.00	0.46
ASTM CA15	0.27	1.00	0.86	0.58	0.39	0.69
17-4PH, H1100	0.46	0.97	1.00	0.13	0.53	0.66
AISI 410	0.41	0.98	0.86	0.59	0.52	0.72

where N represents the normalised value (e.g. N σ_y represents normalised yield stress).

The final ranking of the materials through the MPI formula is tabulated in Table 6 below:

Table 6: Ranked table of the grades of steel with their respective MPI value.

Material	MPI Ranking
50B46	0.903
AISI 4140	0.870
AISI 1080	0.784
AISI 1095	0.782
AISI 1030	0.777
YS550	0.770
AISI 1045	0.761
AISI410	0.724
YS800	0.691
ASTM CA15	0.687
17-4PH, H1100	0.665
AISI 8740	0.549
Maraging Steel 300	0.463

Based on the evaluation of the thirteen materials using the MPI, Low Alloy Steel 50B46 emerged as the highest-performing option, followed by AISI 4140 and Carbon Steel AISI 1080. These materials demonstrated better combined performance across the weighted metrics through the MPI formula, which provided a systematic approach to normalising and ranking materials. AISI 1045, the baseline material used in current DCT shaft designs, ranked seventh, showing that stronger, more cost-effective alternatives exist.

3.3. Mass Calculation of Shafts

The mass of each shaft was calculated through the standard density-volume relationship, shown in Equation 3 below:

$$Mass = \rho \cdot V \qquad \text{Equation 3}$$

- ρ is the density of the material in kg/m^3
- V is the volume of the respective shaft in m^3

The material density was obtained from CES EduPack 2019, while the volume of each shaft was determined using SolidWorks' Mass Properties tool, which provides precise volumetric data based on the CAD geometry.

3.4. Cost Calculation of Shafts

The cost of each shaft was calculated through Equation 4 below:

$$Cost = m \cdot C \qquad \text{Equation 4}$$

- m is the mass of the shaft in kg
- C is the cost per kilogram of the material in $£/kg$

The cost of each shaft can also be calculated through Equation 5 below:

$$Cost = V \cdot C_v \qquad \text{Equation 5}$$

- V is the volume of the shaft in m^3
- C_v is the cost per volume of the material in $£/m^3$

Both shafts were evaluated using these formulae to effectively compare cost against other parameters.

3.5. Finite Element Analysis Setup in Abaqus CAE

FEA was performed using Abaqus CAE's Implicit Solver to simulate the torque applied by the engine through the shaft assembly. The .STEP files were imported as 3D deformable solids. Each shaft was analysed independently, but similar setup procedures were followed. Each analysis used default solver settings and default "Static, General" steps. The "Static, General" step matches the typical static shaft loading analysis using pure torsion. Baseline simulations were conducted to acquire a benchmark for results using predictions of current industry materials and applications. The material used in baseline simulations for both shafts was

AISI1045 Steel due to its standard use in automotive shafts in the automotive industry, and is shown in Table 7 below:

Table 7: Material properties of AISI 1045 steel used in baseline FEA simulations.

AISI1045 Steel	
Youngs Modulus	$205\ GPa$
Shear Modulus	$80\ GPa$
Yield Strength	$530\ MPa$
Fatigue Strength Range at 10^7 Cycles	$287\ MPa - 331\ MPa$
Poisson's Ratio	0.29
Density	$7870\ kg/m^3$
Cost Per Unit Weight	$0.58\ £/kg$

The .STEP files were imported into Abaqus CAE as a 3D deformable solid. The inner input shaft was meshed using free tetrahedral elements of type C3D10 using an average global size of 3 mm, shown in Figure 9:

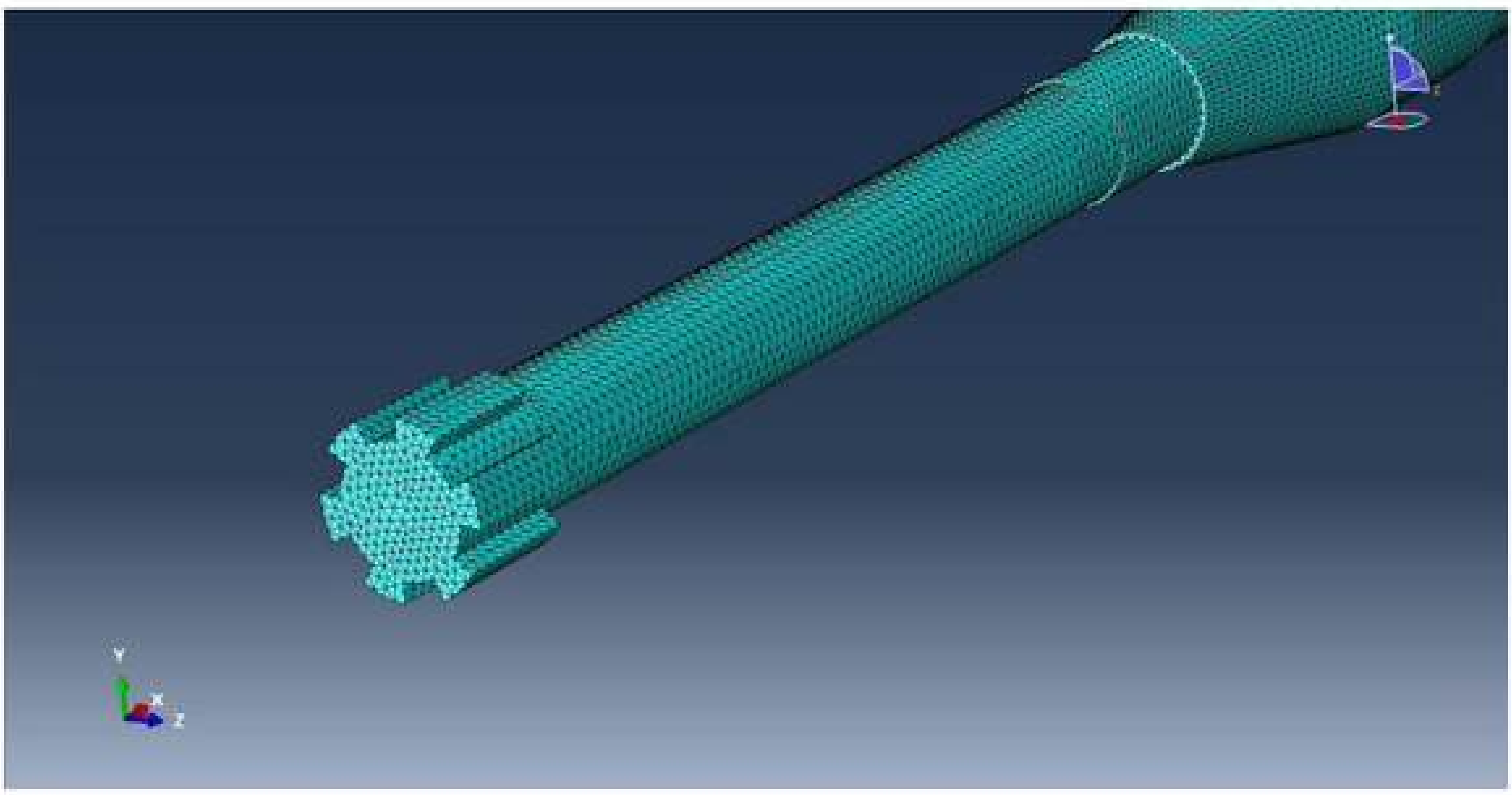

Figure 9: Inner Input Shaft's free tetrahedral mesh of C3D10 elements with an average global size of 3 mm.

The meshed inner shaft was then instanced into an assembly where a reference point was created on the free end of the shaft to allow a load to be applied on the reference point, shown in Figure 10:

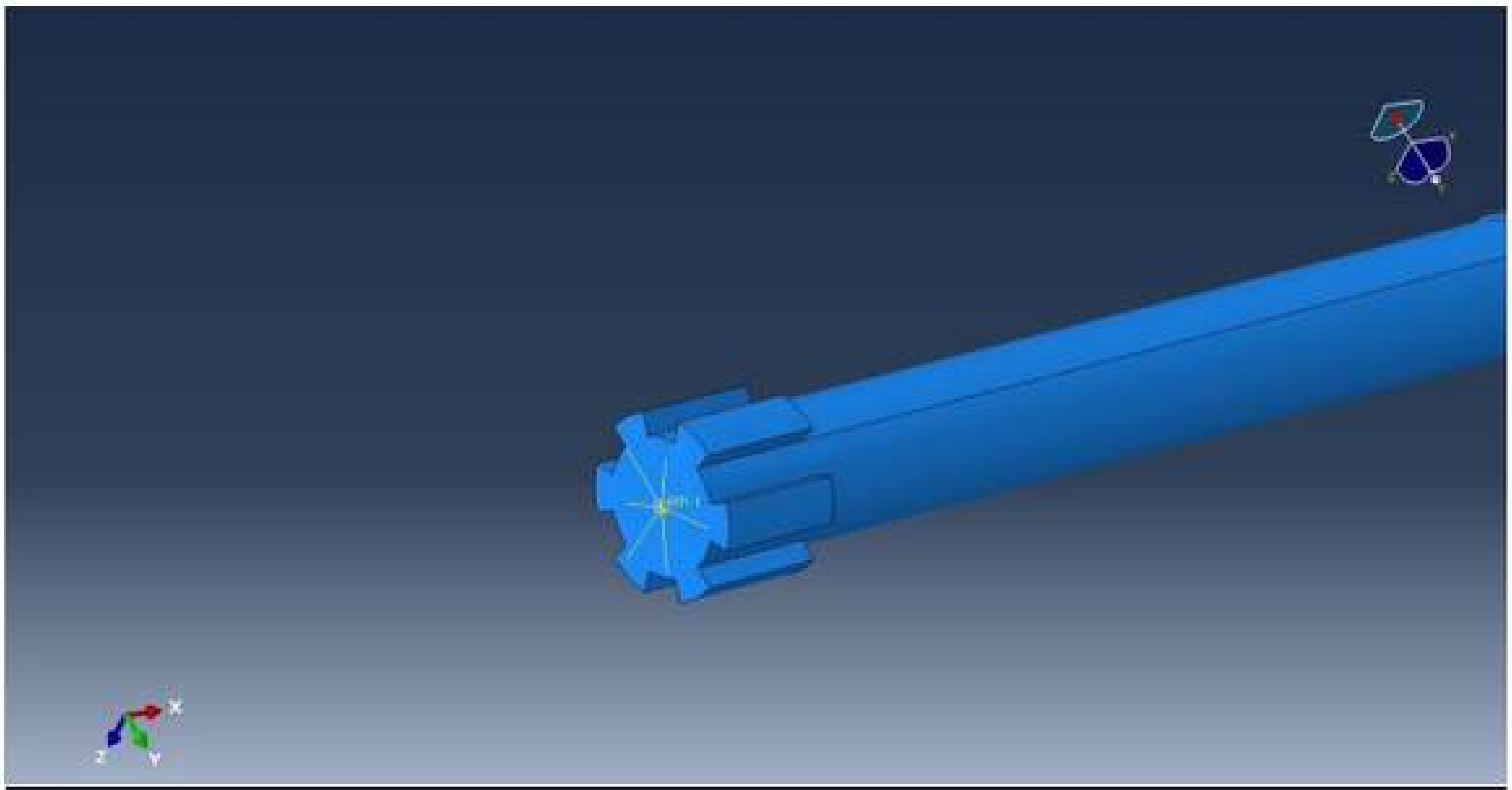

Figure 10: Location of the reference point on the free end of the inner input shaft, used to apply torque through a coupled constraint in the FEA model.

This load would simulate the engine's torque application through the transmission drive shafts. A coupling constraint was applied to the reference point, with the reference point as the master surface and the end face of the shaft near Region 1 as the slave surface. The reference point allows the slave surface to behave consistently with the motion of the reference point. A moment of 420,000 Nmm was applied to the reference point, and a boundary condition was applied to Region 7 of the shaft, corresponding to the location of the first gear, shown in Figure 11:

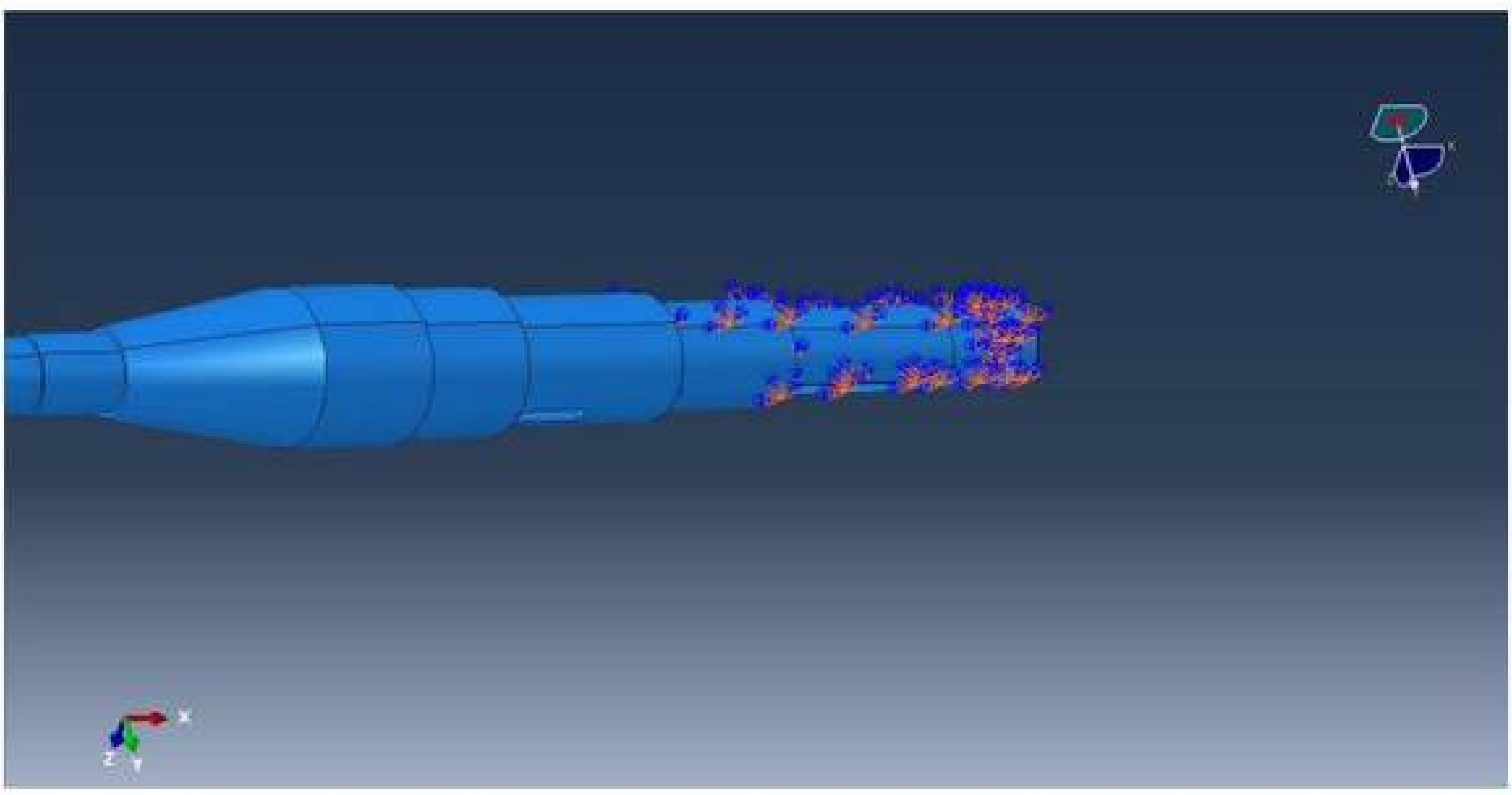

Figure 11: Applied boundary conditions on the inner input shaft, showing the fully constrained Region 7 to simulate fixed support at the location of first gear.

First gear typically transmits the highest torque within the transmission system due to its role in initial acceleration, making it the most appropriate gear through to simulate peak torsional loading. The applied boundary condition represents the gear meshing constraint at the initial onset of torque transmission. A Displacement/Rotation boundary condition was defined at this region, with all translational (U1, U2, U3) and rotational (UR1, UR2, UR3) degrees of freedom fully constrained. This setup effectively simulates a fixed support, preventing any rigid body motion at the gear interface during the simulation.

For the simulation of the outer input shaft, the same procedure was followed as for the inner shaft. The outer shaft was meshed using an average global cell size of 3 mm and free tetrahedral elements of C3D10, as shown in Figure 12:

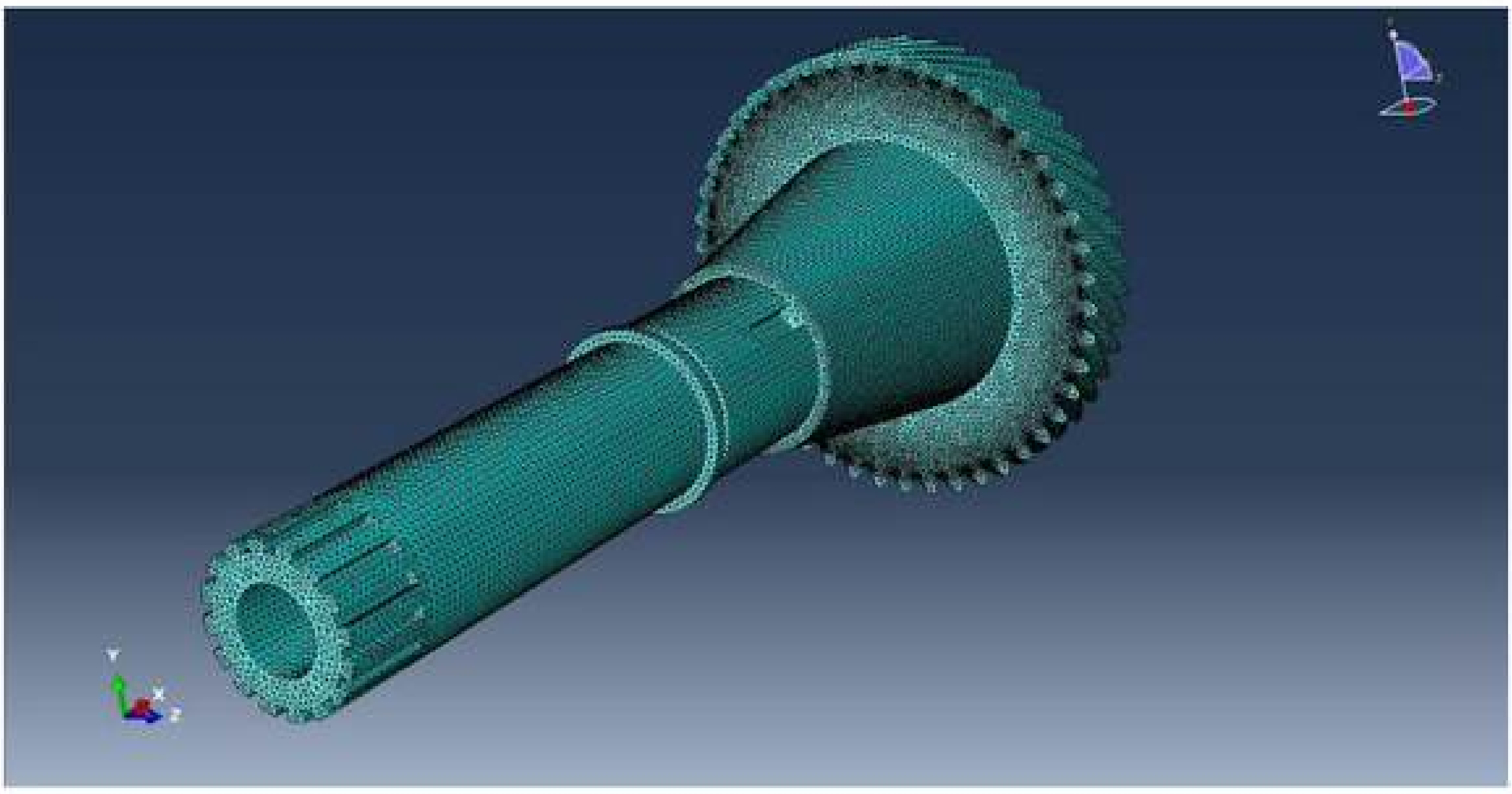

Figure 12: Outer Input Shaft's free tetrahedral mesh of C3D10 elements with an average global size of 3 mm.

The reference point was then applied at the free end of the shaft by Region 1 and constrained with a coupling constraint. Displacement/Rotation boundary conditions were applied, restricting movement in all degrees of freedom (U1, U2, U3, UR1, UR2 and UR3), shown in Figure 13:

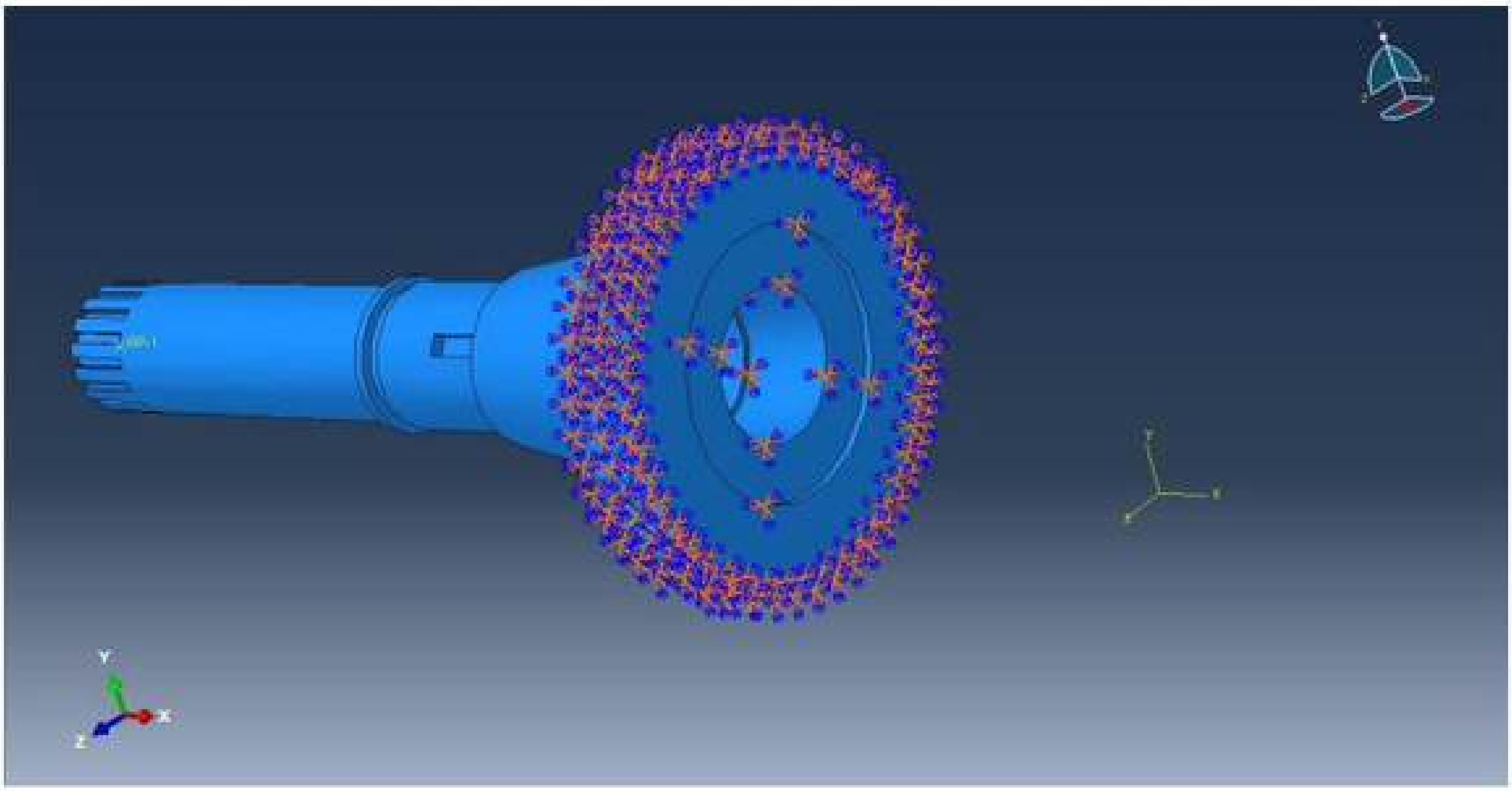

Figure 13: Applied boundary conditions on the outer input shaft, showing the fully constrained Region 3 to simulate fixed support at location of second gear.

No external bending forces, thermal expansion or fatigue were considered for the baseline simulations to simplify the simulations.

4. BASELINE RESULTS & VALIDATION OF THE FEA MODEL

4.1. Baseline Results

This section presents the finite element analysis results for the baseline configuration of the inner and outer DCT input shafts under a 420 Nm torsional load. AISI 1045 steel and a 3 mm global mesh were used for all baseline simulations. Key outputs include torsional shear stress and angle of twist for both shafts.

The weight of the shaft made from AISI1045 Steel was 8.01 kg. Stresses were higher in regions with a smaller diameter. The highest von Mises stress was experienced at the spline root, as shown in Figures 14 and 15 below:

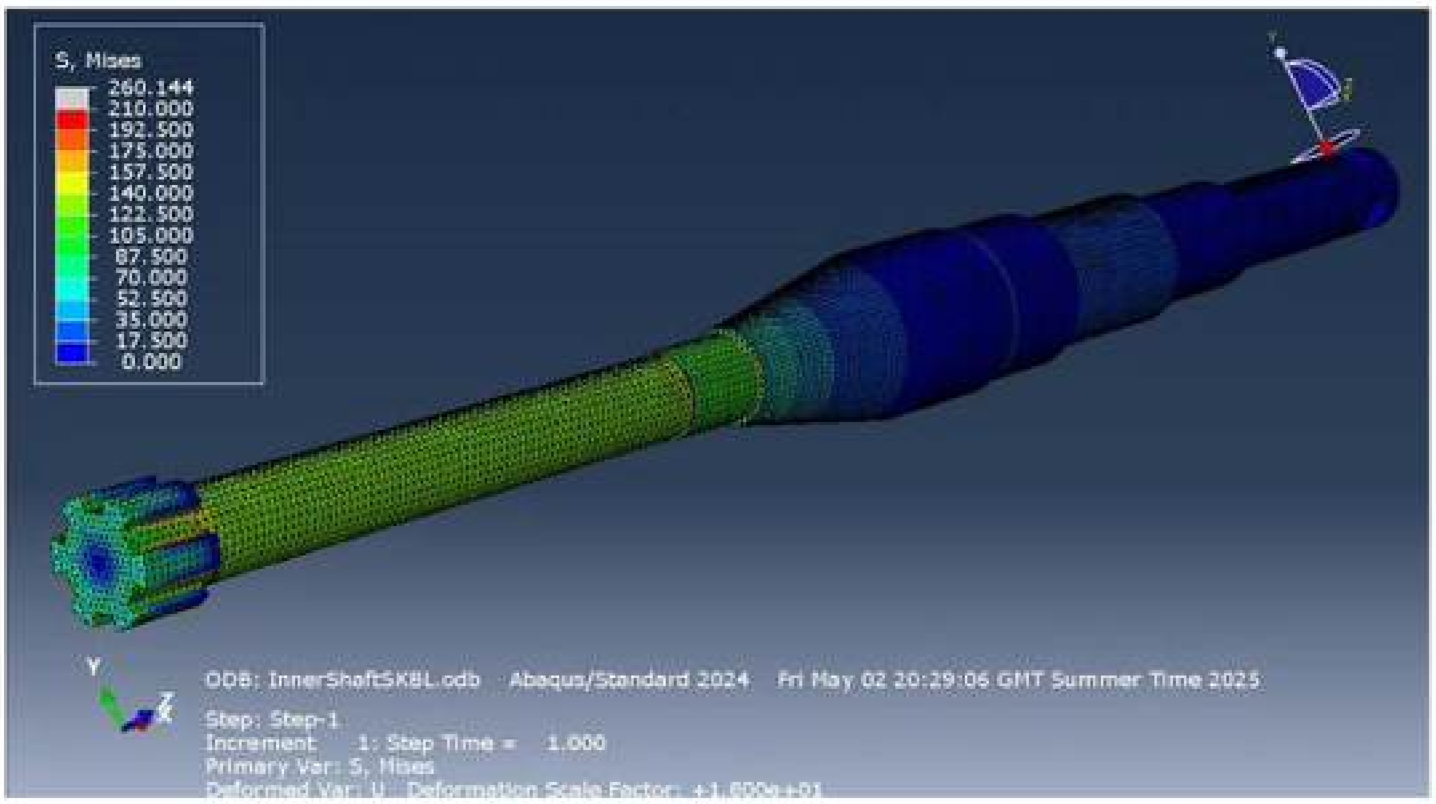

Figure 14: von Mises stress distribution in the inner shaft under 420 Nm torque.

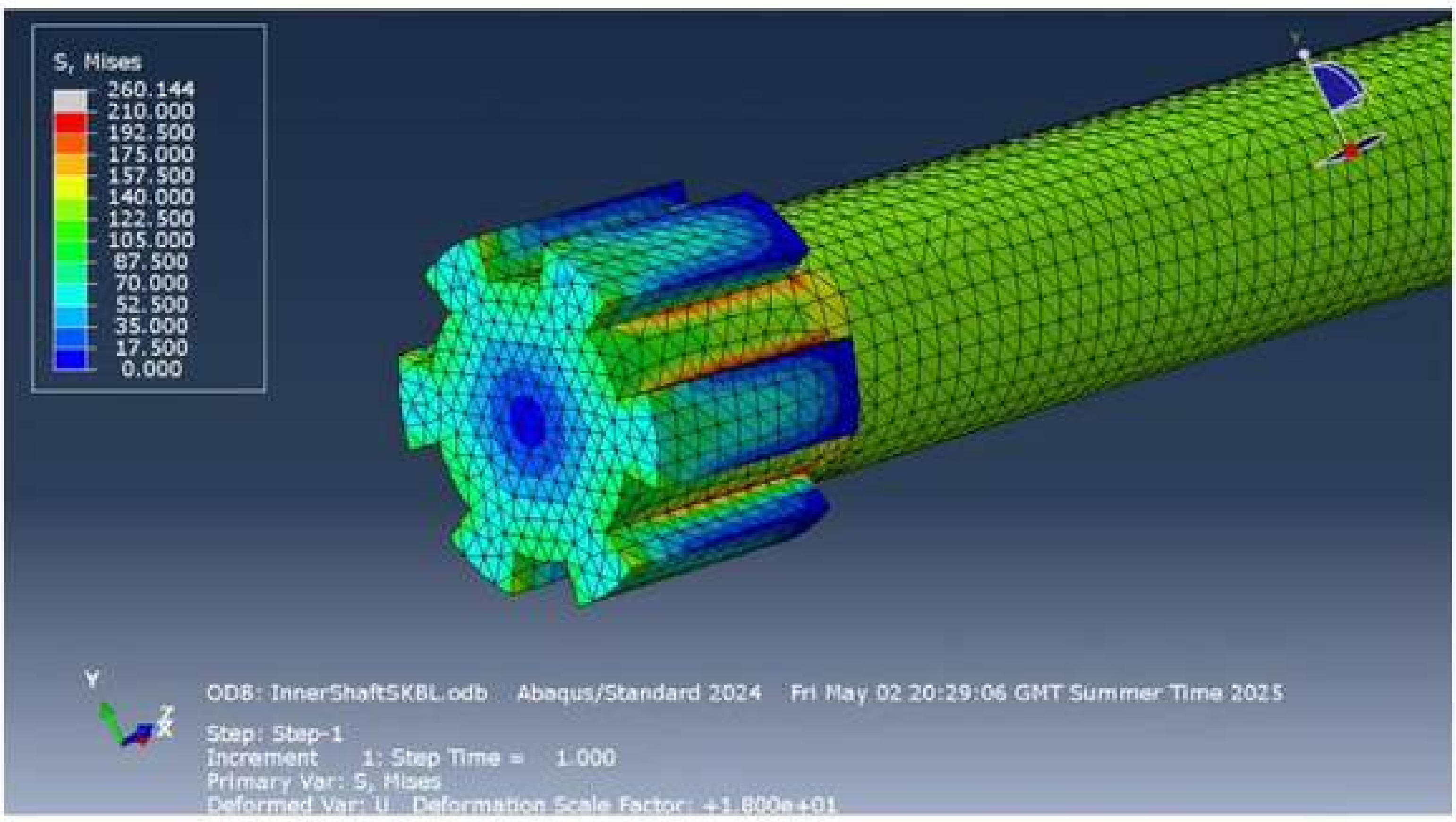

Figure 15: von Mises stress distribution in the inner shaft under 420 Nm torque, highlighting high stress concentrations at the spline roots.

High stress concentrations were expected at sharp diameter transitions as shown in Figure 16:

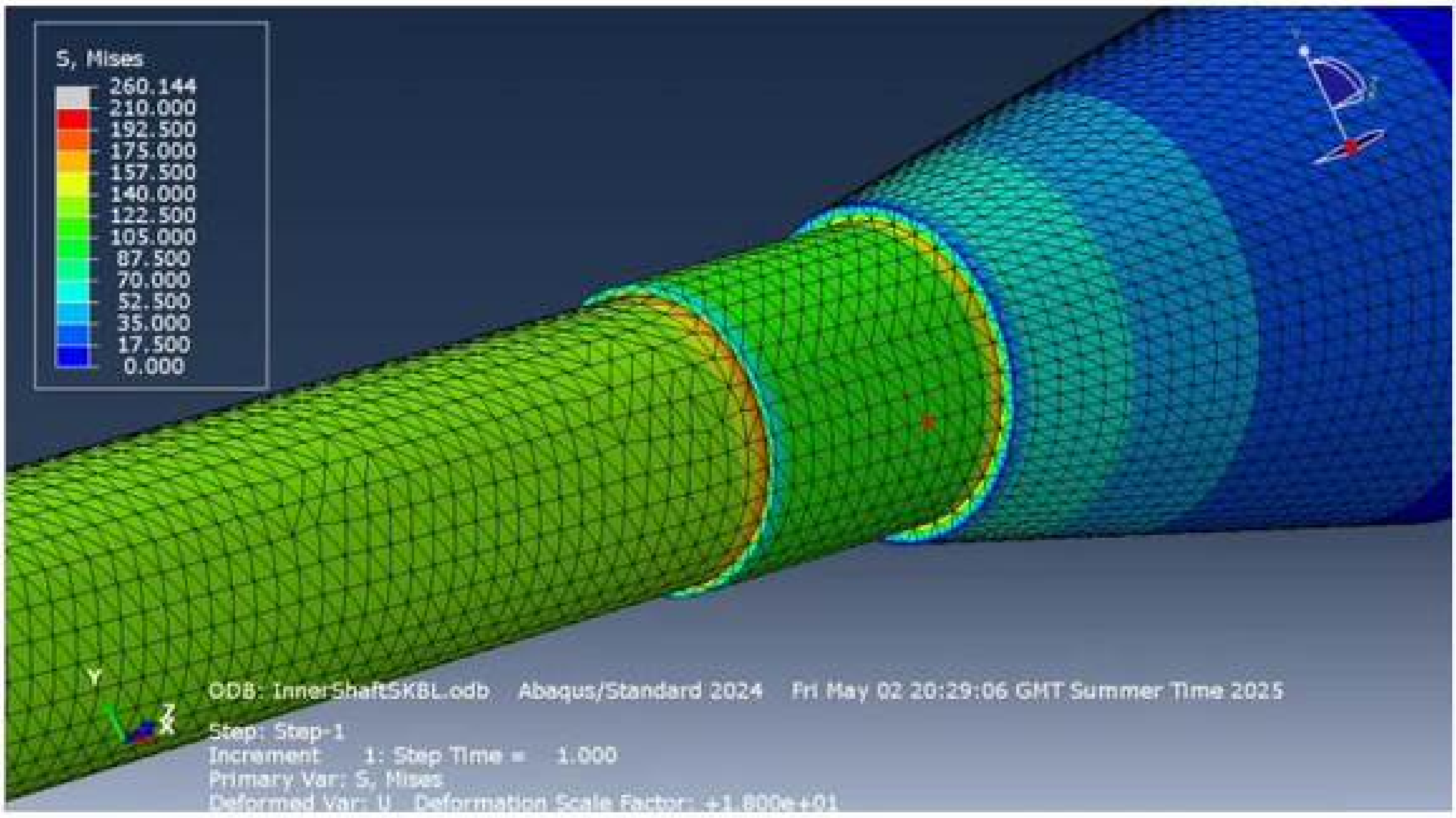

Figure 16: von Mises stress distribution in the inner shaft under 420 Nm torque, highlighting high stress concentrations at the diameter transitions.

The highest angle of twist recorded was at the surface of Region 1, which is the driven end of the shaft. The magnitude of the angle of twist recorded was 0.0206 rad as shown in Figure 17 below:

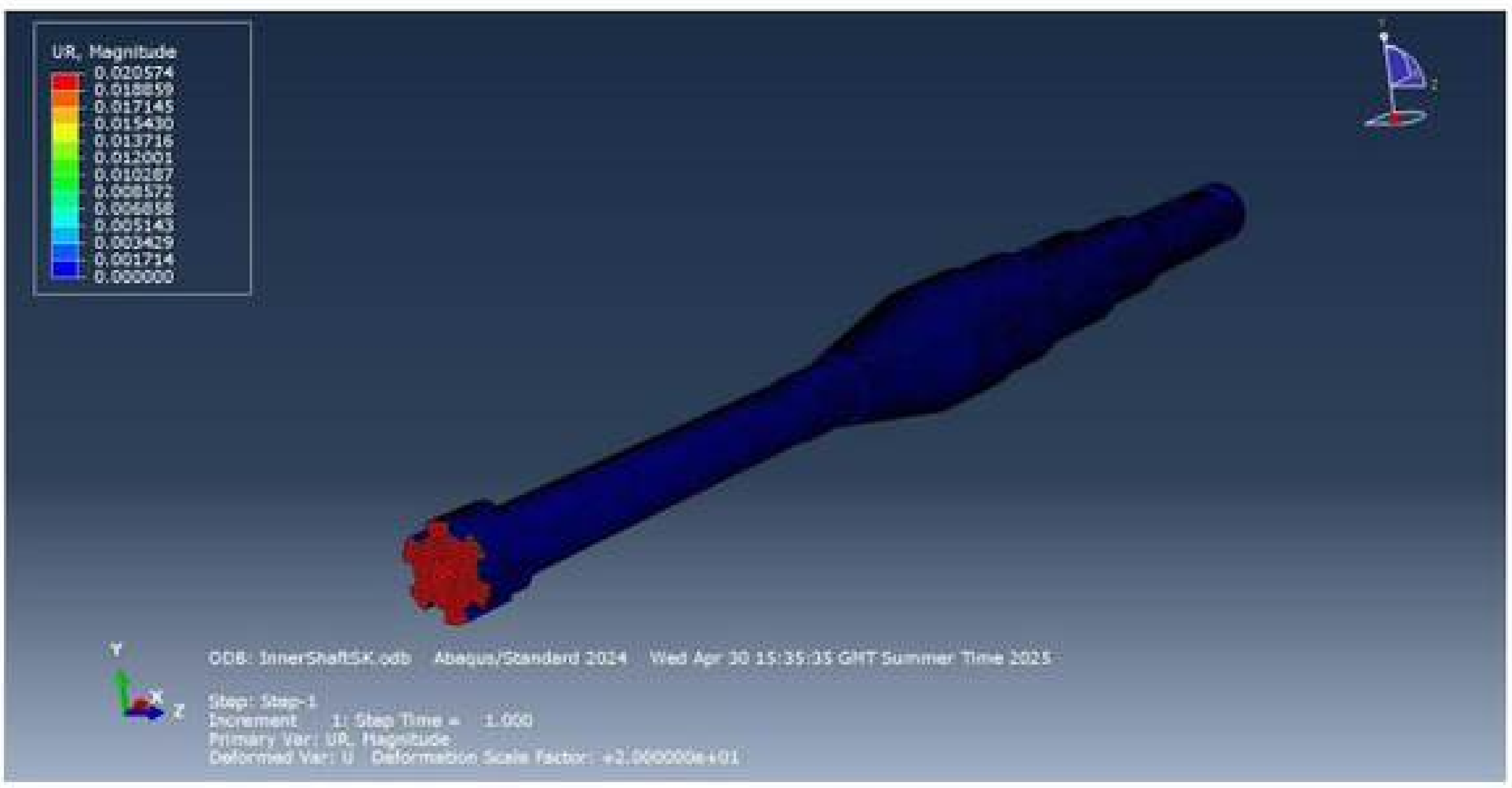

Figure 17: Rotational displacement (UR1) contour of the inner shaft, illustrating the angular deformation profile under applied torsional load.

The highest recorded shear stress was at the spline region of 263 MPa which produces a stress concentration factor of 3.47 compared to a point in Region 1 far away from stress concentrations. The maximum and minimum principal stresses are shown in Figures 18 and 19 below where the highest recorded principal stress was at the spline region:

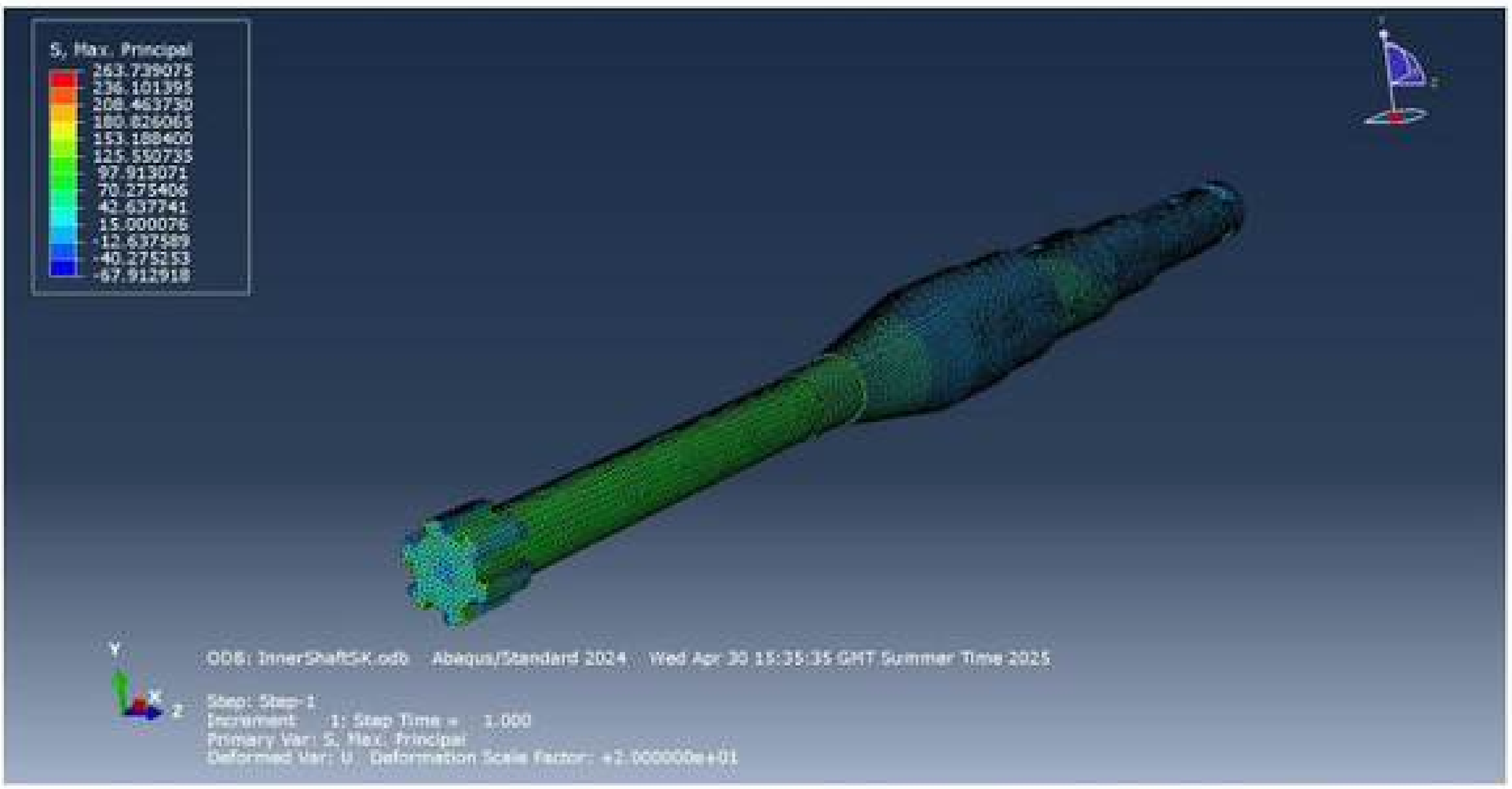

Figure 18: Maximum principal stress (σ_1) distribution in the inner shaft, indicating tensile stress regions aligned with the shaft axis.

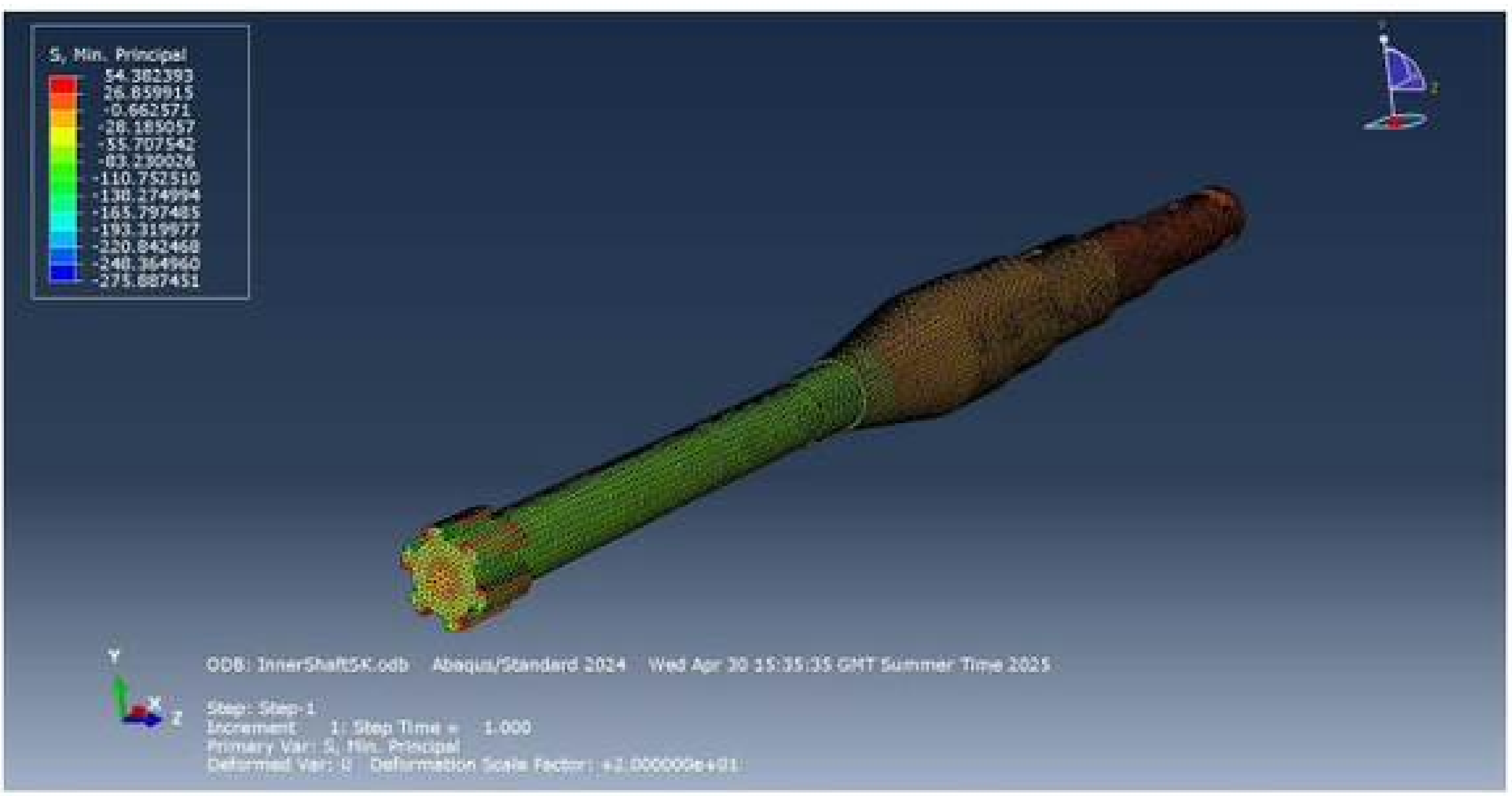

Figure 19: Minimum principal stress (σ_3) distribution in the inner shaft, showing compressive stress regions opposing the primary loading direction.

The weight of the outer shaft made from AISI 1045 Steel was 15.8 kg. The baseline simulation for the outer shaft yielded a lower stress distribution than the input shaft. Nonetheless, the highest recorded von Mises stress was at the spline regions at the point of loading as shown in Figure 20 and 21:

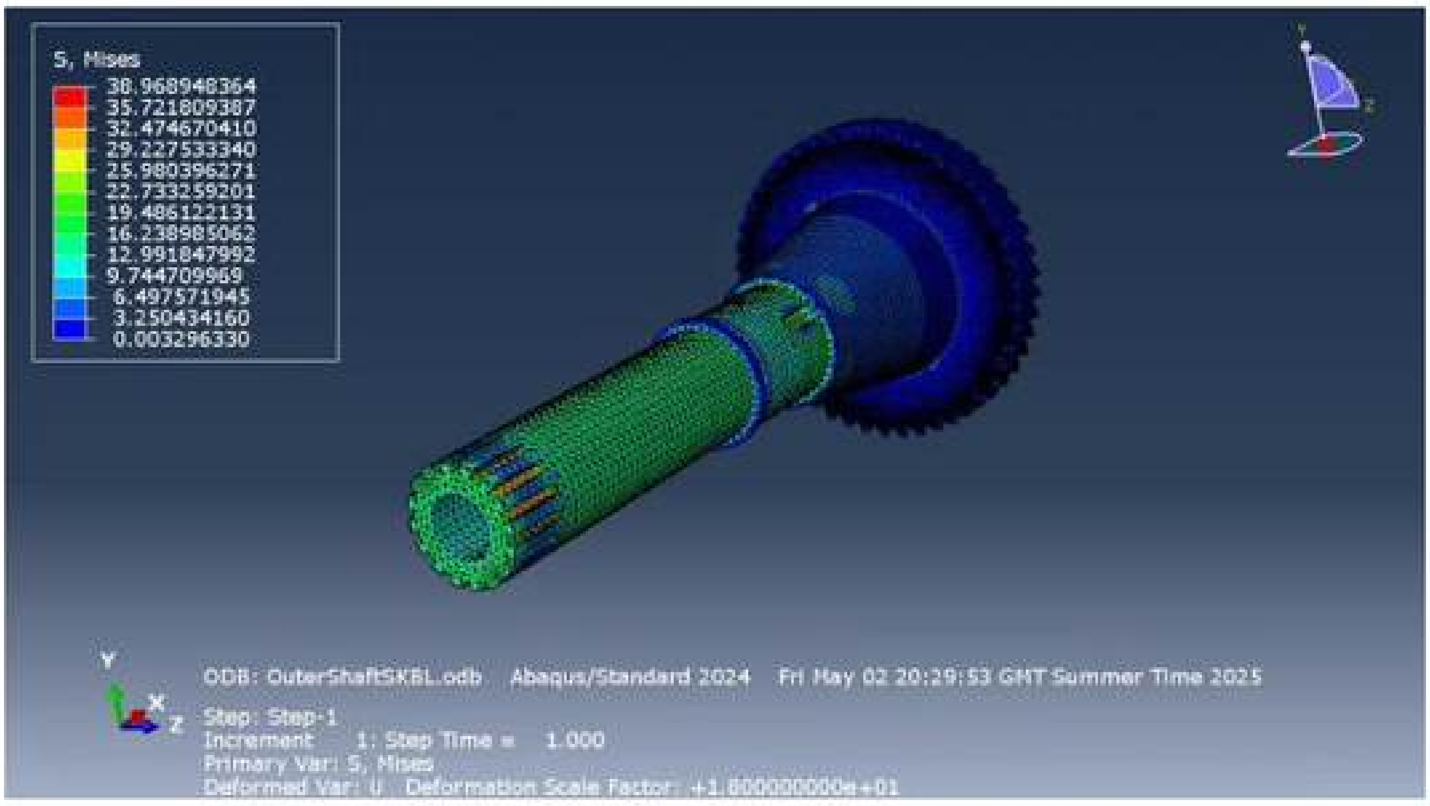

Figure 20: von Mises stress distribution in the outer shaft under 420 Nm torque.

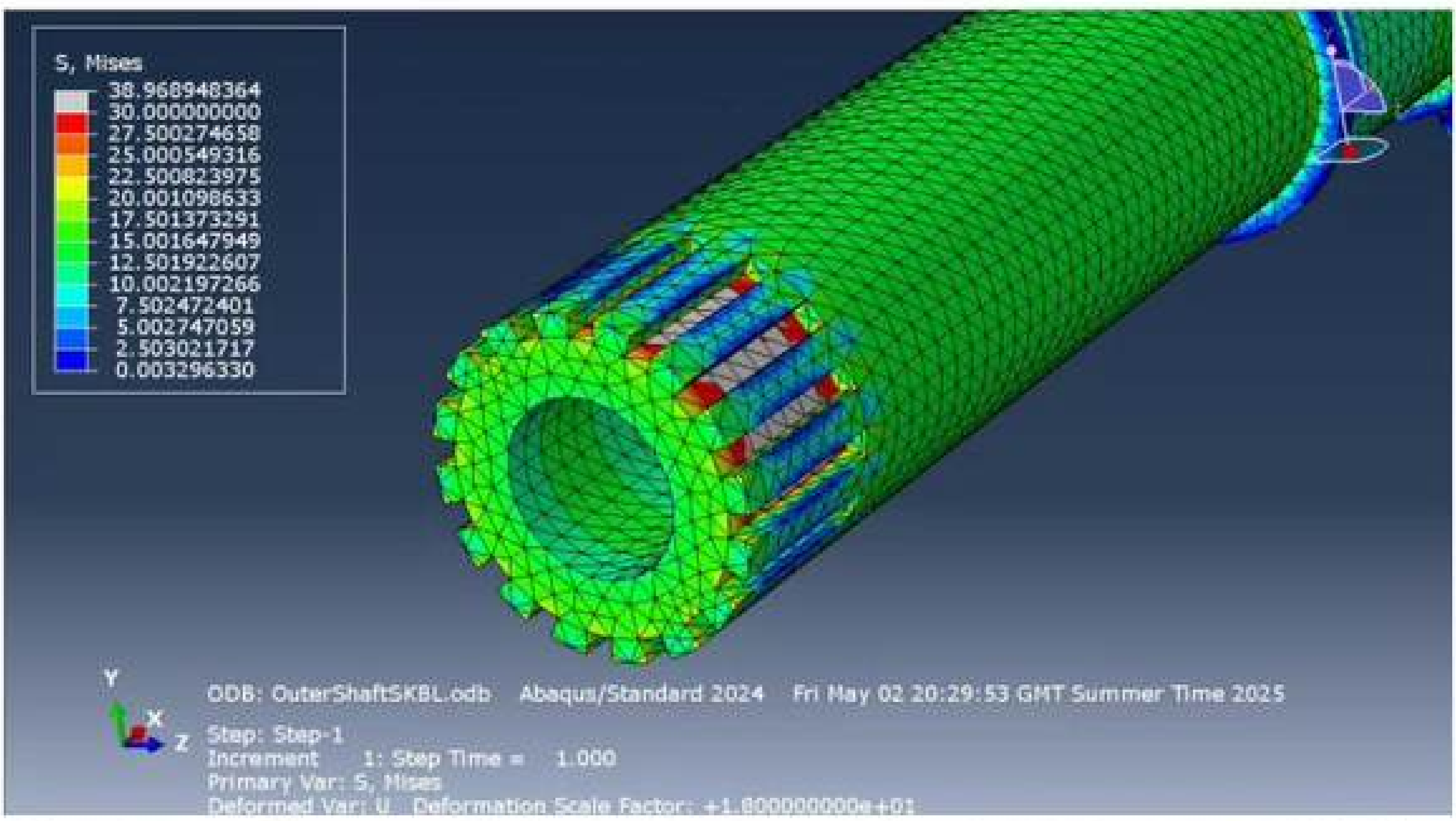

Figure 21: von Mises stress distribution in the outer shaft under 420 Nm torque, highlighting high stress concentrations at the spline root.

High stress concentrations at the sharp edges of the keyway can also be seen in Figure 22:

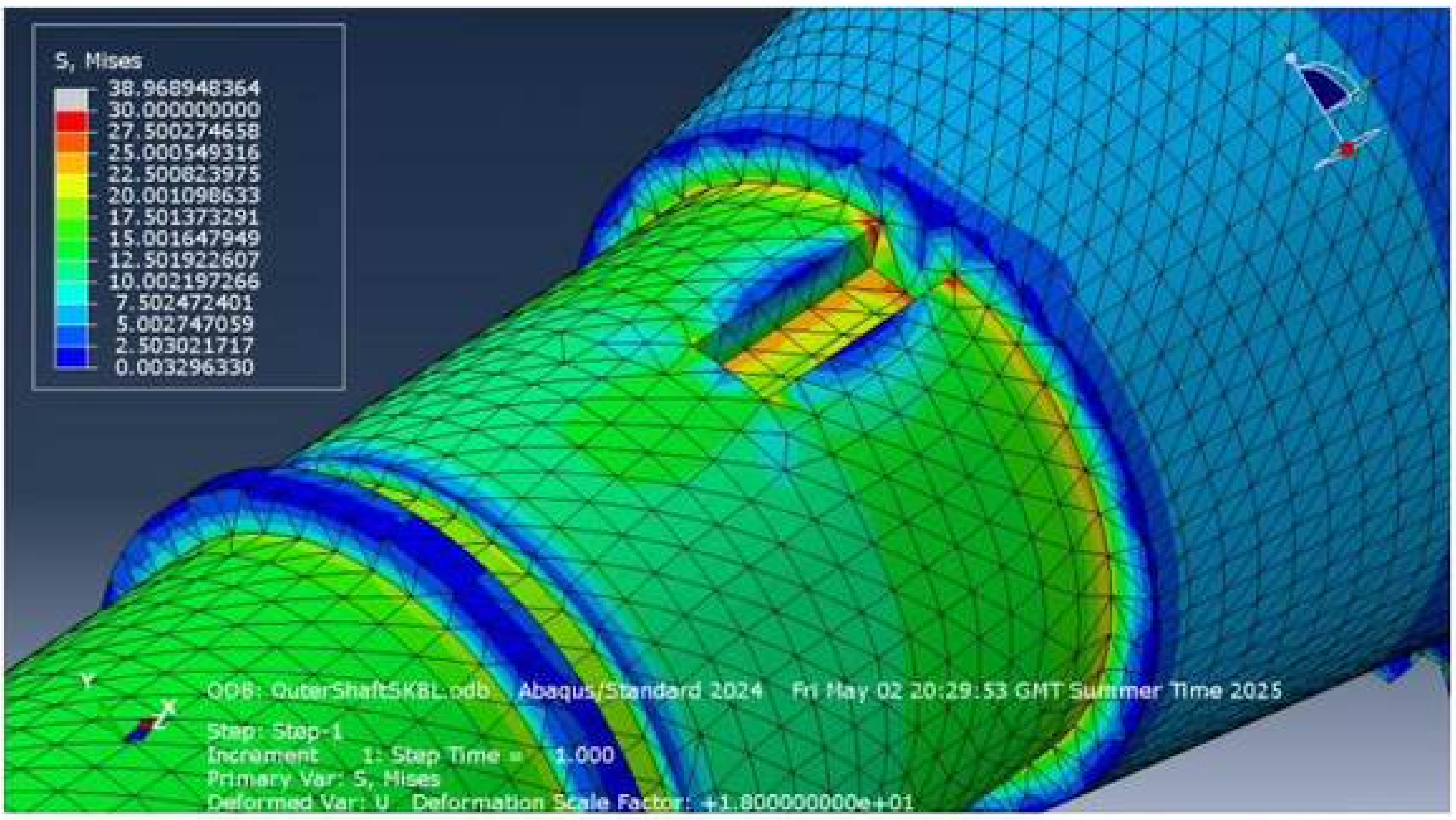

Figure 22: von Mises stress distribution in the outer shaft under 420 Nm torque, highlighting stress concentrations at the keyway slots.

The highest recorded angle of twist at the face of the free end in Region 1 was 0.001 rad which, as expected, is lower than the inner shaft, which is shown in Figure 23:

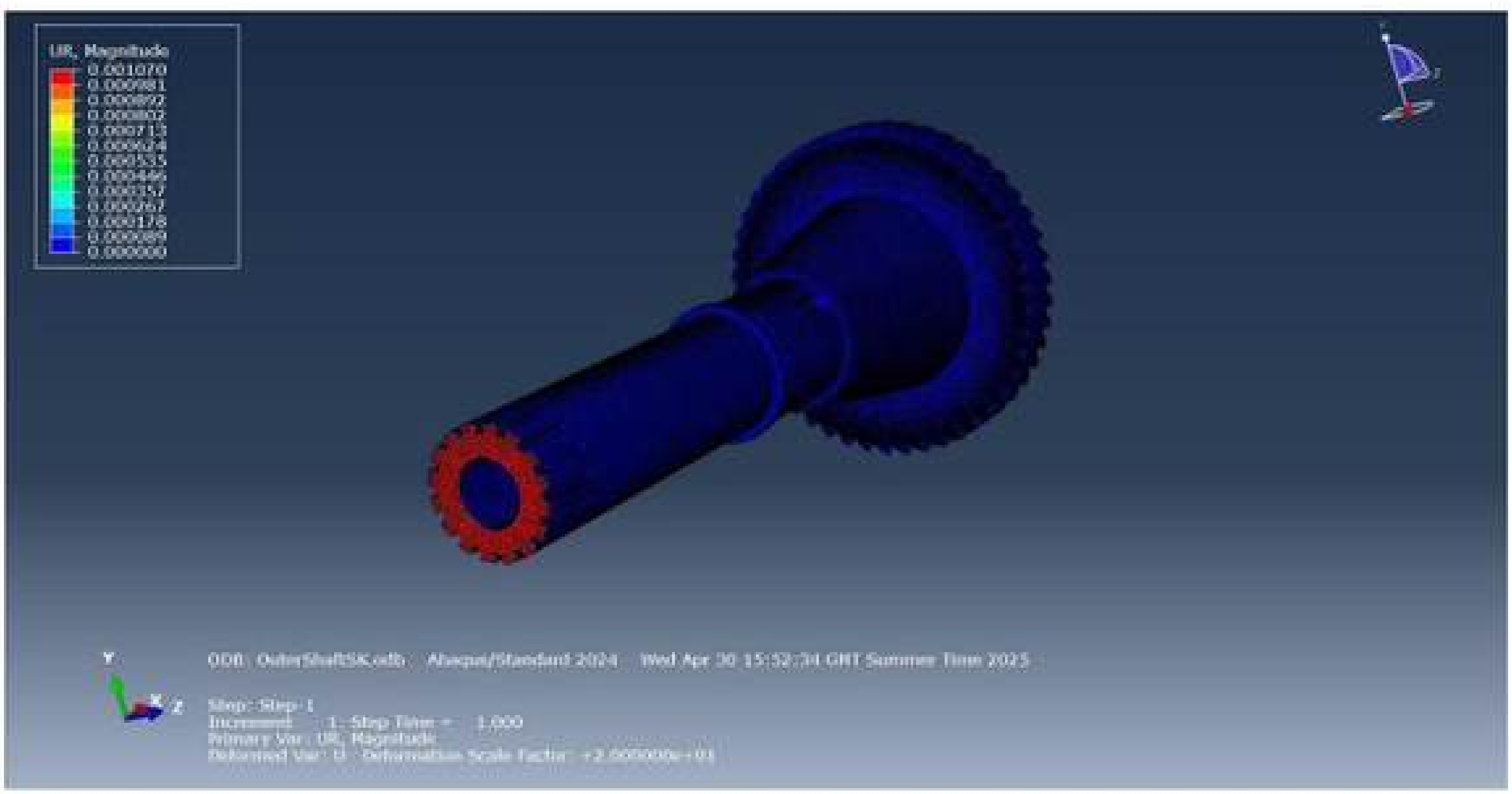

Figure 23: Rotational displacement (UR1) contour of the outer shaft, illustrating the angular deformation profile under applied torsional load.

The highest recorded shear stress was at the spline region of 63.3 MPa which produces a stress concentration factor of 3.64 compared to a point in Region 1 far away from stress concentrations. The maximum and minimum principal stresses are shown in Figures 24 and 25 below where the highest recorded principal stress was at the spline region:

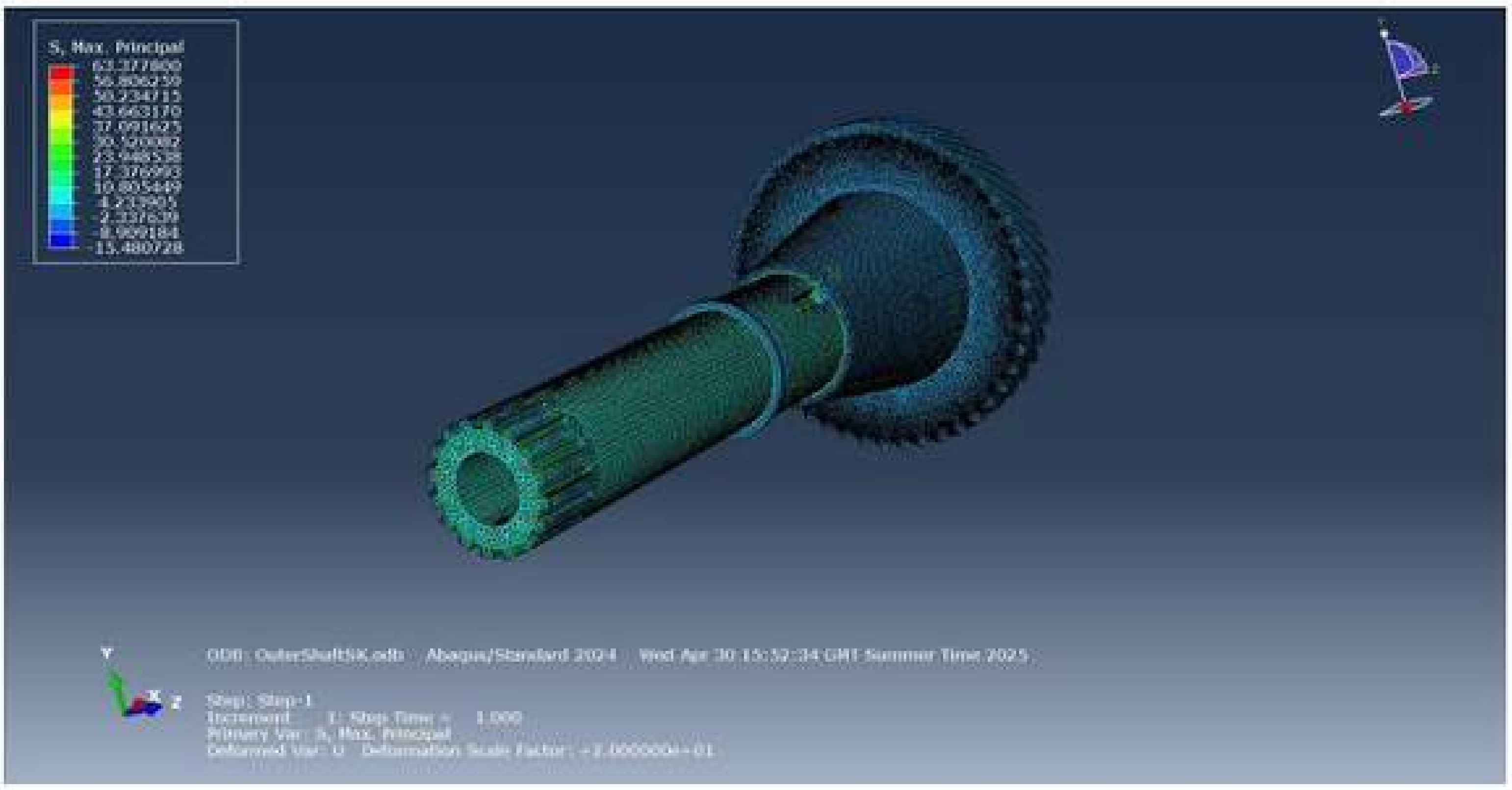

Figure 24: Maximum principal stress (σ_1) distribution in the outer shaft, indicating tensile stress regions aligned with the shaft axis.

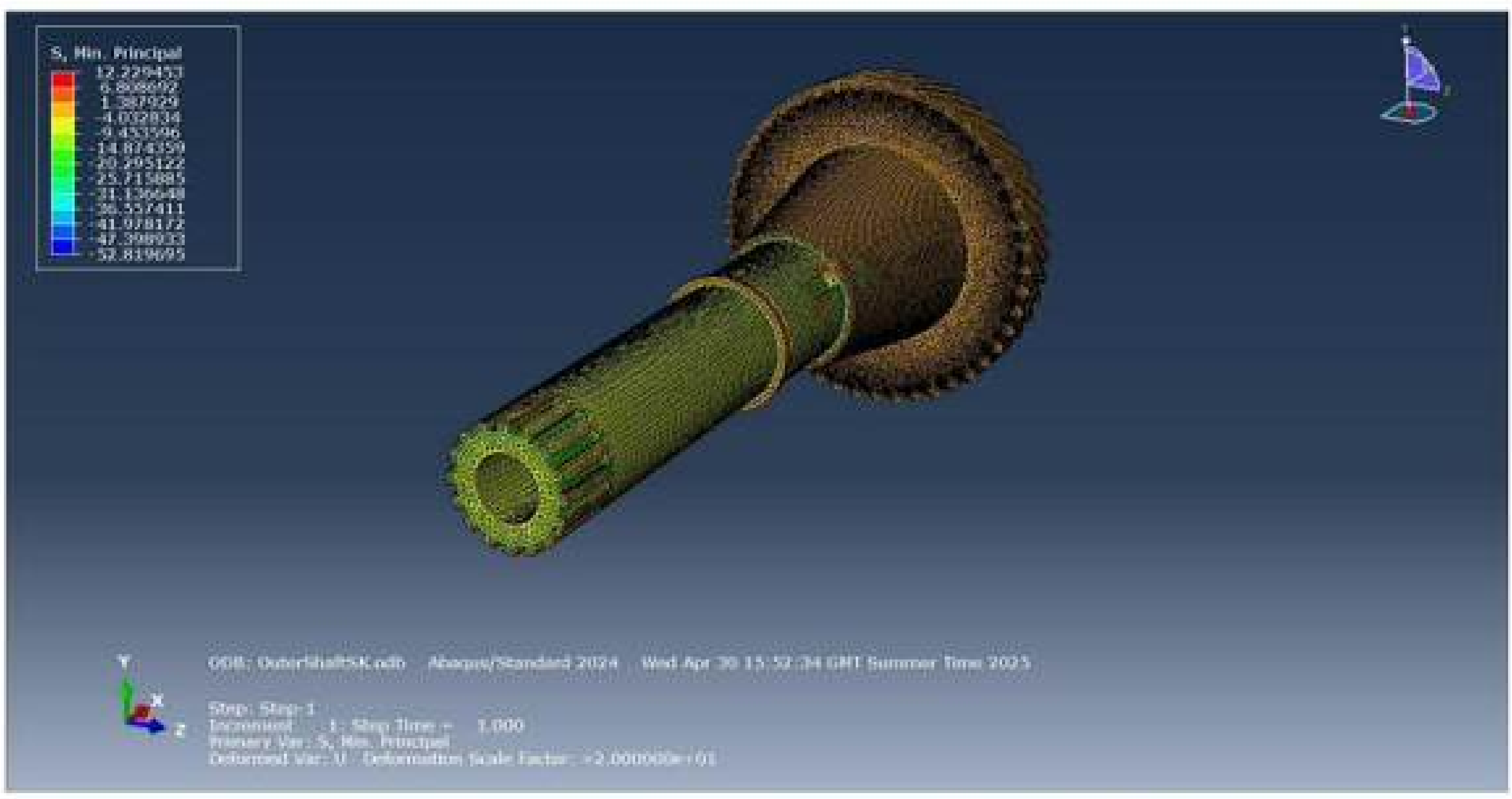

Figure 25: Minimum principal stress (σ_3) distribution in the outer shaft, showing compressive stress regions opposing the primary loading direction.

4.2. Torsional Shear Stress Validation

To validate the accuracy of the torsional shear stress values of different regions of the shafts, results from FEA were compared with classical torsion theory using Equation 6 below:

$$\tau_{THEO} = \frac{TR}{J} \qquad \text{Equation 6}$$

- τ_{THEO} is the theoretical torsional shear stress,
- T is the torque applied to the shaft,
- R is largest radius of the region of the shaft
- J is the polar second moment of area of the region which was calculated using Equation 7 or 8:

$$J\ for\ a\ solid\ shaft = \frac{\pi D^4}{32} \qquad \text{Equation 7}$$

$$J\ for\ a\ hollow\ shaft = \frac{\pi({D_o}^4 - {D_i}^4)}{32} \qquad \text{Equation 8}$$

- D is the diameter of the shaft
- D_o is the outer diameter of a hollow shaft
- D_i is the inner diameter of a hollow shaft

The theoretical shear stress for Regions 2 and 4 for the inner input shaft are tabulated in Table 8 below:

Table 8: Geometric parameters for Regions 2 and 4 of the inner shaft used to calculate the theoretical shear stress.

Inner Shaft Theoretical Shear Stress Calculation

$\boldsymbol{Regions}$	$\boldsymbol{T\ (Nm)}$	$\boldsymbol{Radius\ (m)}$	$\boldsymbol{J\ (x10^{-7}\ m^4)}$	$\boldsymbol{\tau_{THEO}\ (MPa)}$
$Region\ 2$	420	0.016	1.03	65.27
$Region\ 4$	420	0.032	16.47	8.16

Regions 2 and 4 were selected to validate the torsional stress as they have no other geometric features.

The torsional shear stress can be calculated using FEA results through Equation 9 below:

$$\tau_{FEA} = \frac{|\sigma_1 - \sigma_3|}{2} \qquad \text{Equation 9}$$

- τ_{FEA} is the torsional shear stress calculated in FEA,
- σ_1 is the maximum principal stress
- σ_3 is the minimum principal stress.

Equation 9 stems from the Mohr's Circle theory as seen in Figure 26:

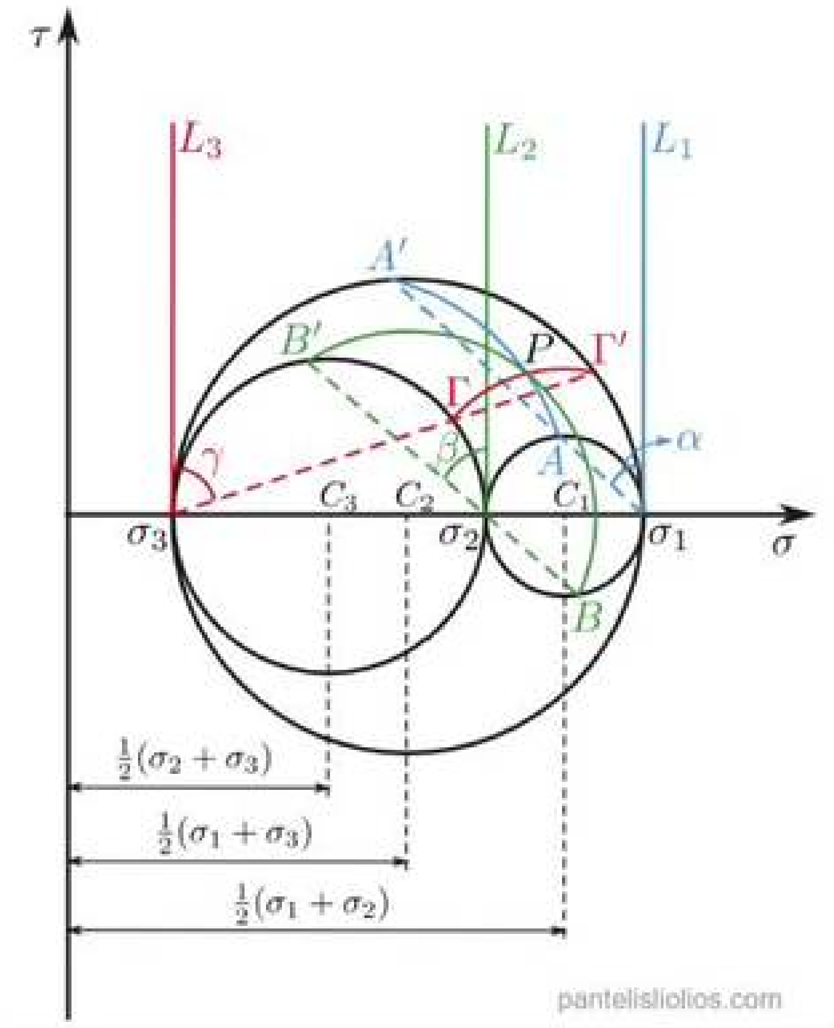

Figure 26: Three-dimensional Mohr's circle diagram illustrating the principal stresses and associated shear stresses [33].

In three-dimensional stress analysis, the maximum shear stress occurs on a plane oriented 45° between the directions of the maximum and minimum principal stresses. This relationship arises because the diameter of the largest Mohr's circle which spans from σ_3 to σ_1 defines the maximum shear stress.

Points on a region far from stress concentrations are probed on Abaqus CAE. The maximum principal stress and minimum principal stress can be acquired and the torsional shear stress seen by the shaft in the FEA simulation can be calculated. The calculation of the torsional shear stress acquired through Abaqus CAE for Regions 2 and 4 for the inner input shaft are tabulated in Table 9 below:

Table 9: Maximum and Minimum principal stresses probed from the baseline simulation for the inner shaft on Abaqus CAE, used to calculate the shear stress.

Inner Shaft FEA Shear Stress Calculation

$Regions$	$\sigma_1\ MPa$	$\sigma_3\ (MPa)$	$\tau_{FEA}\ (MPa)$
$Region\ 2$	63.11	-63.00	63.06
$Region\ 4$	7.98	-7.97	7.98

The theoretical torsional shear stress and shear stress calculated through FEA can be compared to validate the FEA model shown in Table 10 below:

Table 10: Comparison of theoretical shear stress and shear stress extracted from the baseline simulation for the inner shaft.

Shear Stress Comparison for Inner Shaft

$Regions$	$\tau_{THEO}\ (MPa)$	$\tau_{FEA}\ (MPa)$	$Percentage\ Error$
$Region\ 2$	65.27	63.06	-3.39%
$Region\ 4$	8.16	7.98	-2.21%

The same calculations can be done for the outer shaft shown in Tables 11, 12 and 13:

Table 11: Geometric parameters for Regions 1 and 2 of the outer shaft extracted using the SolidWorks measure tool, used to calculate the theoretical shear stress.

Outer Shaft Theoretical Shear Stress Calculation

$Regions$	$T\ (Nm)$	$Radius(m)$	$J\ (x10^{-7}\ m^4)$	$\tau_{THEO}\ (MPa)$
$Region\ 1$	420	0.0325	16.21	8.42
$Region\ 2$	420	0.035	22.26	6.60

Table 12: Maximum and Minimum principal stresses probed from the baseline simulation for the outer shaft shaft on Abaqus CAE, used to calculate the shear stress.

Outer Shaft FEA Shear Stress Calculation

$Regions$	$\sigma_1\ MPa$	$\sigma_3\ (MPa)$	$\tau_{FEA}\ (MPa)$
$Region\ 1$	8.21	-8.22	8.21
$Region\ 2$	7.08	-7.06	7.07

Table 13: Comparison of theoretical shear stress and shear stress extracted from the baseline simulation for the outer shaft.

Shear Stress Comparison for Outer Shaft

$Regions$	$\tau_{THEO}\,(MPa)$	$\tau_{FEA}\,(MPa)$	$Percentage\ Error$
$Region\ 1$	8.42	8.21	-2.49%
$Region\ 2$	6.60	7.07	7.12%

The comparison between theoretical shear stress and the shear stress values derived from Abaqus CAE shows strong agreement, with differences within an acceptable tolerance. This confirms that the FEA model accurately captures the torsional loading response. The percentage errors can be attributed to the simplification of the theoretical calculations where geometry such as splines and keyways were neglected. The validation supports the reliability of the stress outputs from the FEA model and forms a solid foundation for evaluating material and geometric modifications in later simulations.

4.3. Angle Of Twist Validation

To further validate the FEA simulations through theoretical calculations, the angle of twist was calculated for each shaft to compare against FEA values. The calculations were calculated through Equation 10 below:

$$\theta_{THEO}=\sum\frac{TL}{JG} \qquad \text{Equation 10}$$

- $\theta_{THEORETICAL}$ is the angle of twist.
- T is the torque applied to the shaft.
- L is the length of each section.
- J is the polar second moment of area for each section.
- G is the material's shear modulus.

The angle of twist was calculated for each region of a shaft separately and summed to give the angle of twist for the whole shaft which is tabulated in Table 14 below for the inner shaft:

Table 14: Recorded geometric values for calculating the angle of twist for the inner shaft.

Angle Of Twist Calculation for Inner Shaft

$Regions$	$Diameter$ (m)	$Length$ (m)	$J\ (x10^{-7}\ m^4)$	$G\ (GPa)$	$T\ (Nm)$	θ_{THEO} (rad)
$Region\ 1$	0.03	0.260	0.795	80	420	0.0172
$Region\ 2$	0.032	0.0355	1.03	80	420	0.0018
$Region\ 3$	0.0498	0.08036	6.04	80	420	0.0007
$Region\ 4$	0.064	0.0485	16.47	80	420	0.0002
$Region\ 5$	0.060	0.04564	12.72	80	420	0.0002
$Region\ 6$	0.052	0.075	7.18	80	420	0.0005
$Region\ 7$	0.044	0.05717	3.68	80	420	0.0008
$Region\ 8$	0.036	0.080	1.65	80	420	0.0025
$Region\ 9$	0.035	0.042	1.47	80	420	0.0015
		$Sum\ of\ Angle\ of\ Twist\ (rad)$				0.0254
		$Angle\ of\ Twist\ Extracted\ from\ FEA\ (rad)$				0.0221
		$Percentage\ Error$				-12.99%

Several geometric assumptions were made to simplify the theoretical calculations. The spline in Region 1 was neglected, and Region 1 was modelled as a smooth circular shaft. In Region 3, an average diameter was used to represent the tapered geometry. Additionally, the keyways in Regions 6 and 8 and the slot in Region 9 were neglected; these sections were treated as solid cylinders. These simplifications allowed for analytical estimation of the polar moment of area and angle of twist for each region.

The same method of calculating the angle of twist can be applied to the output shaft shown in Table 15:

Table 15: Recorded geometric values for calculating the angle of twist for the outer shaft.

Angle Of Twist Calculation for Outer Shaft							
$Regions$	$Diameter_i$ (m)	$Diameter_o$ (m)	$Length$ (m)	$J\ (x10^{-7}\ m^4)$	$G\ (GPa)$	$T\ (Nm)$	θ_{THEO} (rad)
$Region\ 1$	0.034	0.065	0.22	16.21	80	420	0.000712
$Region\ 2$	0.034	0.07	0.0655	22.26	80	420	0.000154
$Region\ 3$	0.034	0.1016	0.14229	103.13	80	420	0.000072
$Sum\ of\ Angle\ of\ Twist\ (rad)$							0.000939
$Angle\ of\ Twist\ Extracted\ from\ FEA\ (rad)$							0.001069
$Percentage\ Error$							13.95%

The angle of twist results derived from FEA closely match those calculated theoretically, further validating the model's global deformation behaviour. Given the small magnitude of deformation for both shafts, the percentage error indicates that the applied boundary conditions, mesh quality, and material assignment were appropriate. Since angle of twist is a function of the full shaft geometry and not just local regions, this agreement confirms that the model accurately represents overall shaft stiffness under torsional loading.

4.4. Mesh Convergence Study

A mesh convergence study was conducted to validate the numerical accuracy of the finite element model and determine an optimal element size that balances precision with computational efficiency. The study was performed independently on the inner and outer input shafts using second-order tetrahedral elements of C3D10M in Abaqus/CAE. Global mesh seed sizes of 15 mm, 10 mm, 8 mm, 6 mm, 4 mm, and 2 mm were tested. The angular displacement about the shaft axis (UR1) at the free reference point was used as the primary convergence metric, as it represents the global torsional deformation and is less sensitive to local stress singularities. The results of the mesh sensitivity analysis are shown below in Tables 16 and 17 for the inner and outer shafts, respectively:

Table 16: Recorded values for calculating of angle of twist convergence for the inner shaft.

Inner Shaft		
$Cell\ Size\ (mm)$	$No. of\ Elements$	$Maximum\ \theta$
15	3895	0.023136266
10	9080	0.022744862
8	16984	0.022250244
6	33166	0.022168092
4	90566	0.0220509
2	519456	0.021978641

Table 17: Recorded values for calculating of angle of twist convergence for the outer shaft.

Outer Shaft		
$Cell\ Size\ (mm)$	$No. of\ Elements$	θ
15	26846	0.001056159
10	44007	0.001062047
8	66665	0.001063865
6	166652	0.001065917
4	385582	0.001068365
2	1928166	0.001069121

A plot of the refinement of the angle of twist against the number of elements for the inner shaft is plotted in Figure 27 to visualise the mesh convergence:

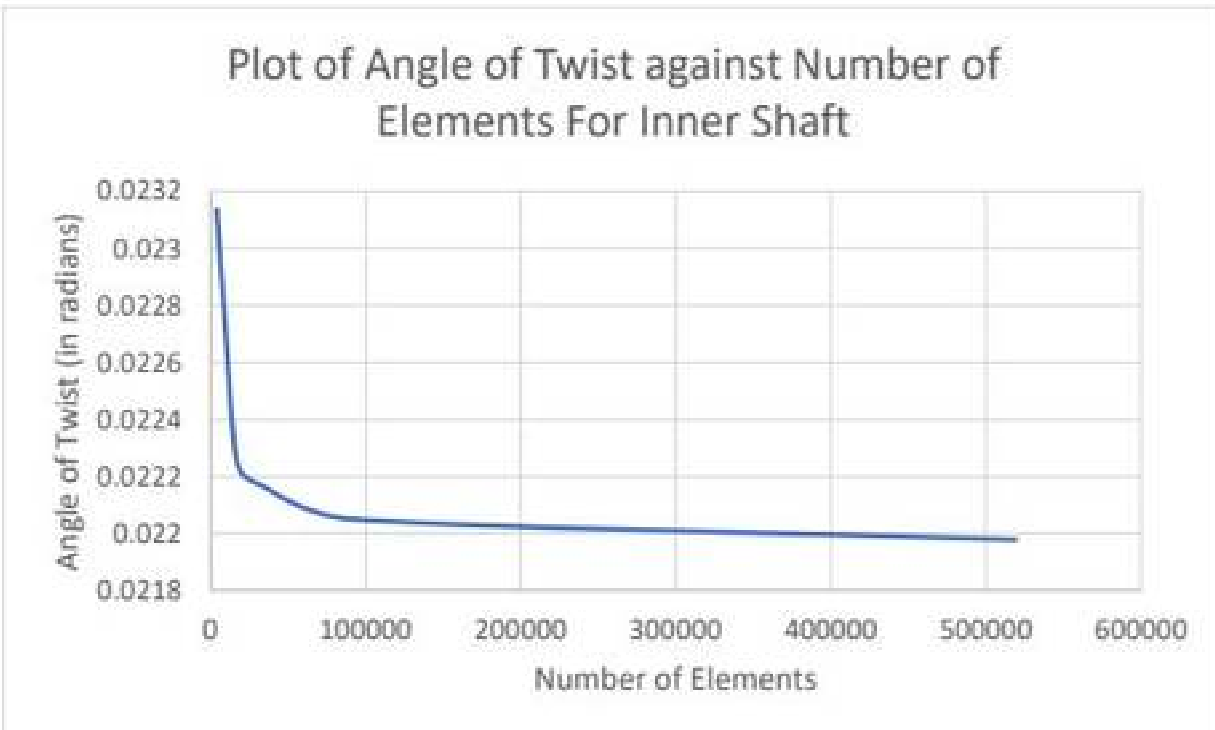

Figure 27: Plot of change in angle of twist for increasing number of elements for the inner shaft.

As seen in Figure 27, the UR1 value stabilises for the inner shaft as the mesh is refined, with the most significant variation occurring between the 15 mm mesh and the 6 mm mesh. The relative change in twist between the 4 mm and 2 mm meshes is off by a small magnitude, indicating convergence. The 2 mm mesh gives marginally more accurate results. The optimal mesh size selected, which best balances accuracy and computation time, is a 4 mm mesh size, is a 4 mm mesh size.

Another plot of the refinement of the angle of twist against the number of elements for the outer shaft is plotted in Figure 28 to visualise the mesh convergence:

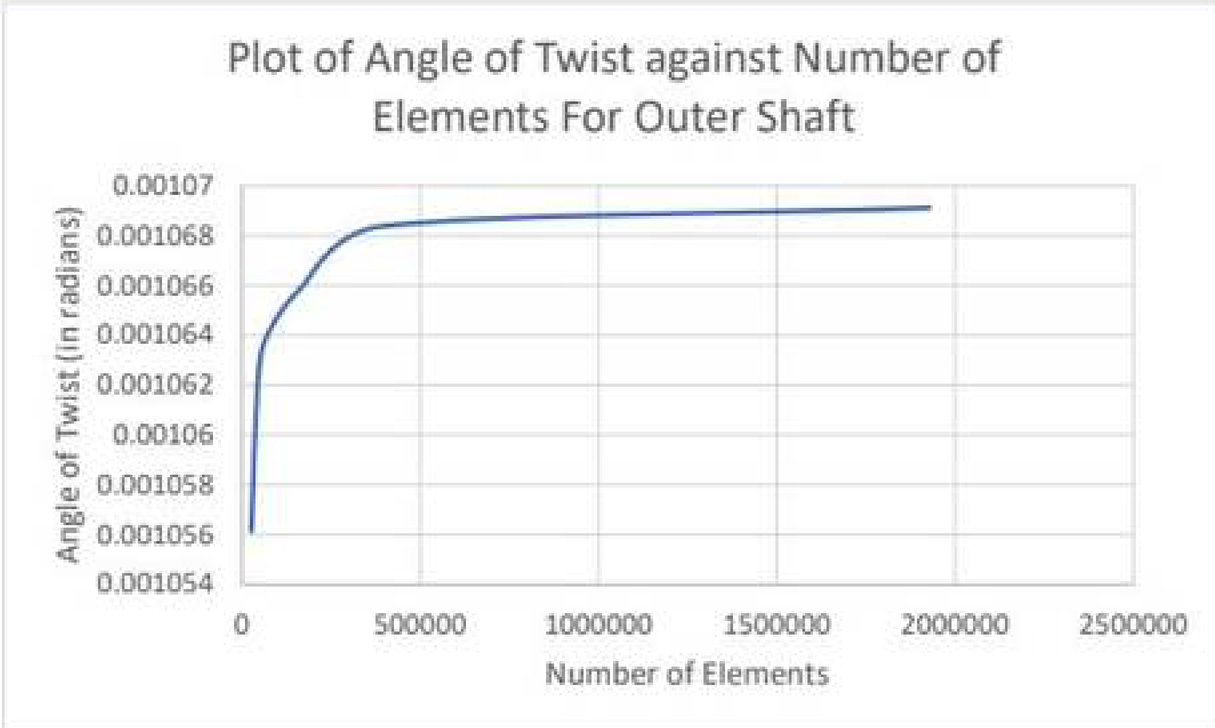

Figure 28: Plot of change in angle of twist for increasing number of elements for the outer shaft.

For the outer shaft, while mesh refinement finer than 8 mm showed continued improvements in angular twist convergence, the increase in element count beyond this point resulted in high computational time. The huge increase in elements between 6 mm and 2 mm significantly increased simulation runtime, with diminishing returns in accuracy. Therefore, refinement beyond 8 mm was deemed impractical within the Project's time constraints, and an 8 mm mesh was used to ensure consistency with the inner shaft while maintaining manageable simulation durations.

4.5. Factor of Safety

The factor of safety (FoS) of a material was calculated to assess whether the inner and outer input shafts could withstand the applied torsional loads without yielding, based on material strength and simulated stress results. The FoS is a dimensionless measure defined in Equation 11 as:

$$Factor\ of\ Safety = \frac{\sigma_y}{\sigma_{max}} \qquad \text{Equation 11}$$

- σ_y is the yield strength of the material
- σ_{max} is the maximum equivalent von Mises stress observed from FEA

The von Mises stress was extracted from the Abaqus output using probe tools for both the inner and outer shafts. Maximum stress values were identified in critical regions such as spline roots, diameter transitions, and keyways which features known to display high stress concentrations.

Material yield strengths were taken from standard datasheets corresponding to the specific treatment and condition of each material tested. The FoS was calculated individually for each shaft and material variant, including the baseline simulation and other simulation cases involving alternative steels and alloys.

A minimum acceptable FoS of 2 was considered for static loading conditions, consistent with conventional mechanical design guidelines. Values above this threshold suggest overdesign and potential opportunities for material or geometry optimisation, while values below indicate insufficient strength or risk of yielding under operational loads.

4.6. Summary of Baseline Input Shafts Results

A summary of the results from the baseline simulations are tabulated in Table 18:

Table 18: Summary of key results from baseline simulations for the inner and outer shaft.

	Inner Shaft	**Outer Shaft**
Average Global Mesh Size (mm)	3 mm	3 mm
Material	AISI 1045	AISI 1045
Mass (kg)	8.007	15.799
Cost Per Shaft $(£)$	4.65	9.17
Maximum von Mises Stress (MPa)	260.143	38.969
Factor of Safety	2.03	13.6
Angle of Twist (rad)	0.02057	0.00107

5. RESULTS

5.1. Mass Comparison

Table 19 below presents the calculated mass of the inner and outer shafts for each material, along with the percentage change relative to the baseline material, AISI 1045, which is highlighted in bold. Table 19 is ranked with the lightest shafts through to the heaviest shafts:

Table 19: Shaft weights and percentage difference in mass compared to AISI 1045.

Material	Weight of Inner Shaft (kg)	Weight of Outer Shaft (kg)	$\Delta\%$ Difference From Baseline Simulations
ASTM CA15	7.76	15.32	-3.06
AISI 410	7.91	15.60	-1.27
AISI 1030	8.01	15.80	0.00
AISI 1095	8.01	15.80	0.00
AISI 1080	8.01	15.80	0.00
YS800	8.01	15.80	0.00
YS550	8.01	15.80	0.00
50B46	8.01	15.80	0.00
AISI 4140	8.01	15.80	0.00
AISIS 8740	8.01	15.80	0.00
AISI 1045	***8.01***	***15.80***	***0.00***
17-4PH, H1100	8.02	15.82	0.13
Maraging Steel 300	8.08	15.93	0.85

Figure 29 presents a visual comparison of the mass performance of all materials evaluated in the study. The bar chart illustrates the percentage difference in total shaft weight for each material relative to the baseline, AISI 1045, highlighting candidates with potential for weight reduction:

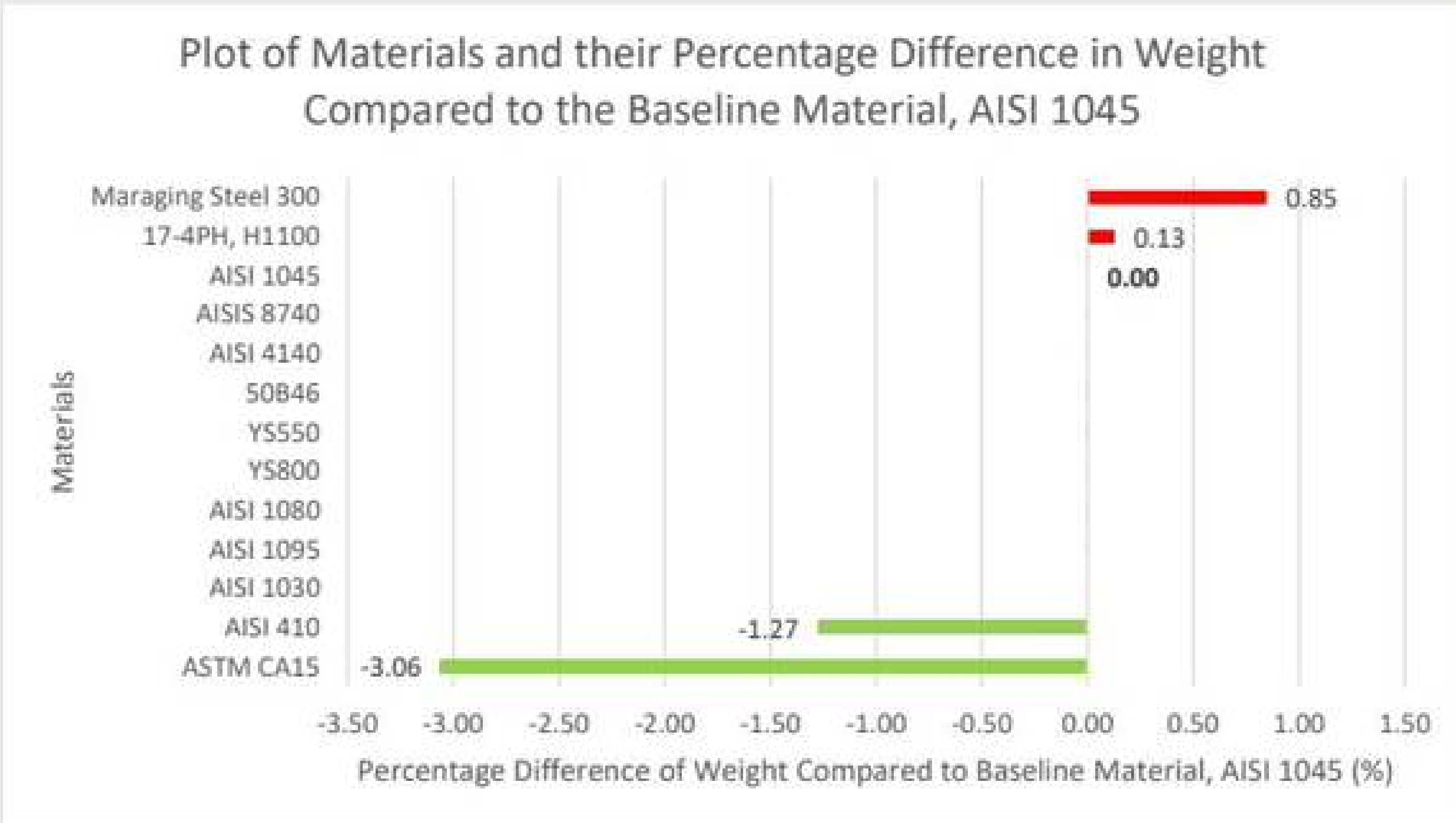

Figure 29: Percentage difference in shaft weight for each material relative to AISI 1045. Positive values indicate heavier materials; negative values indicate lighter alternatives.

5.2. Cost Comparison

The cost of an inner and outer input shaft was calculated for each prospective material and is detailed in Table 20 below. The materials in Table 20 are ordered from lowest to highest total cost, with the most cost-effective options listed at the top:

Table 20: Shaft costs and percentage difference in cost compared to AISI 1045.

Material	Cost of Inner Shaft	Cost of Outer Shaft	Δ% Difference From Baseline Material
17-4PH, H1100	3.53	6.96	-24.12
YS550	4.59	9.06	-1.32
AISI 1095	4.59	9.06	-1.32
AISI 1045	***4.65***	***9.18***	***0.00***
AISI 1080	4.65	9.18	0.01
AISI 1030	4.65	9.18	0.01
50B46	4.71	9.30	1.35
AISI 4140	5.05	9.97	8.59
AISIS 8740	5.45	10.75	17.08
ASTM CA15	7.67	15.14	64.91
AISI 410	7.76	15.32	66.88
YS800	8.95	17.66	92.43
Maraging Steel 300	226.22	446.34	4763.22

The table highlights the cost variations across materials, which directly impact material feasibility for large-scale production. Figure 30 below presents a plot of the material and its respective % difference of cost from AISI 1045. Maraging Steel 300 was excluded from Figure 30 due to its significantly higher price relative to the other materials. Including it would have disproportionately skewed the visual scale, making comparisons between more practical alternatives less clear. However, its cost data is still considered in the tabulated results and final material evaluation:

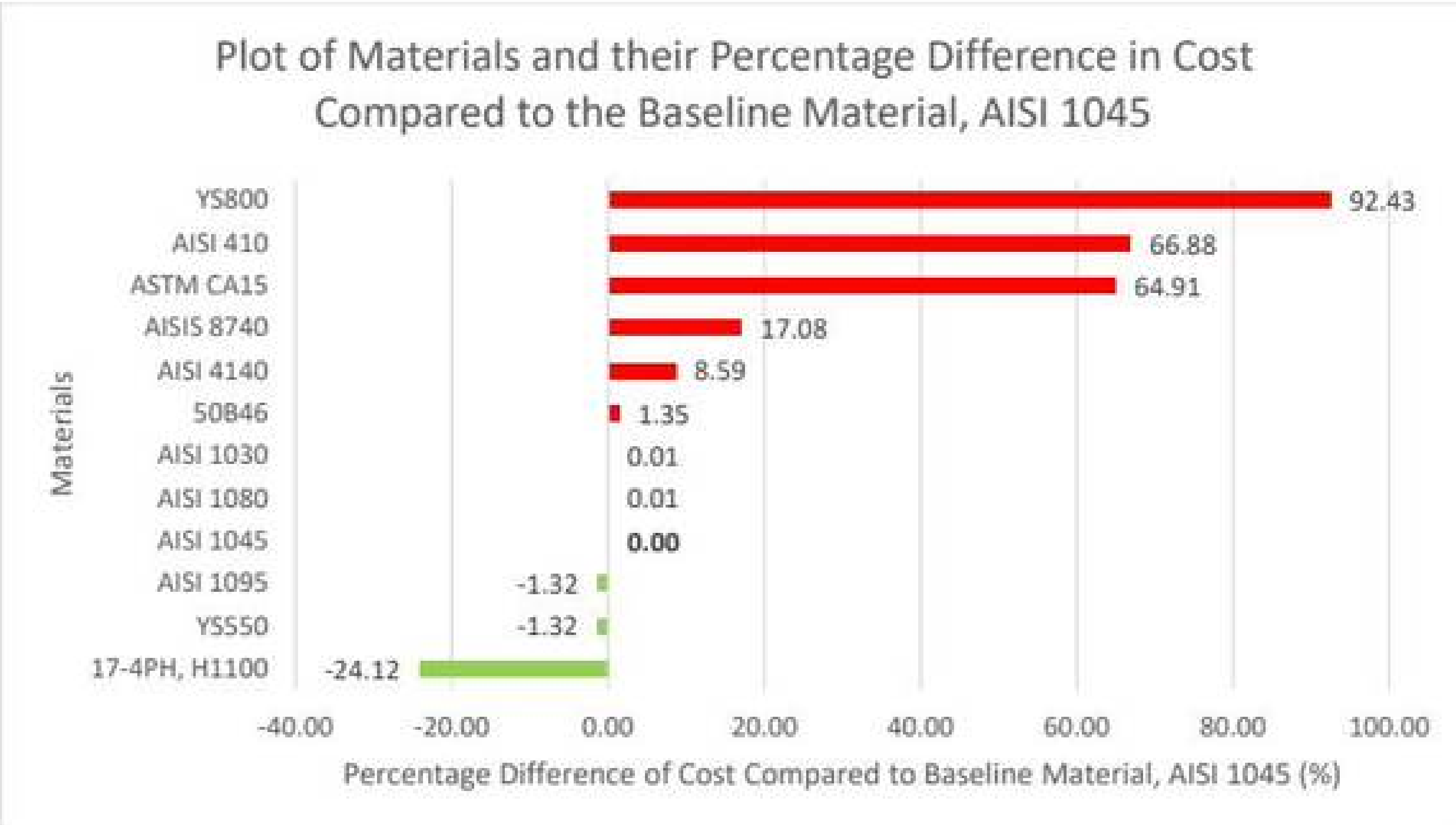

Figure 30: Percentage difference in total shaft cost per material compared to AISI 1045. Maraging Steel 300 was excluded due to extreme cost skew.

5.3. Stress Comparison

The von Mises stress results and calculated FoS for both the inner and outer shafts under identical loading conditions are tabulated below in Table 21:

Table 21: Maximum von Mises stress and corresponding factor of safety values for inner and outer shafts across all material variants.

Material	Inner Shaft		Outer Shaft	
	Maximum von Mises Stress	Factor of Safety	Maximum von Mises Stress	Factor of Safety
Maraging Steel 300	260.60	7.29	39.07	48.66
50B46	260.14	6.23	38.97	41.57
AISI 4140	260.14	5.51	38.97	36.76
AISI 8740	260.14	5.22	38.97	34.84
17-4PH, H1100	259.88	3.35	38.91	22.37
YS800	260.33	3.07	39.01	20.51
AISI 410	259.96	2.98	38.93	19.92
YS550	260.33	2.30	39.01	15.38
AISI 1080	260.14	2.26	38.97	15.08
AISI 1030	260.14	2.23	38.97	14.88
AISI 1095	260.14	2.12	38.97	14.18
AISI 1045	***260.14***	***2.04***	***38.97***	***13.60***
ASTM CA15	259.97	1.98	38.93	13.23

Figures 31 and 32 presents a visual comparison of the highest recorded FoS for all materials for the inner and outer shaft, respectively. The bar chart illustrates which materials provide a better structural performance than AISI 1045 in green:

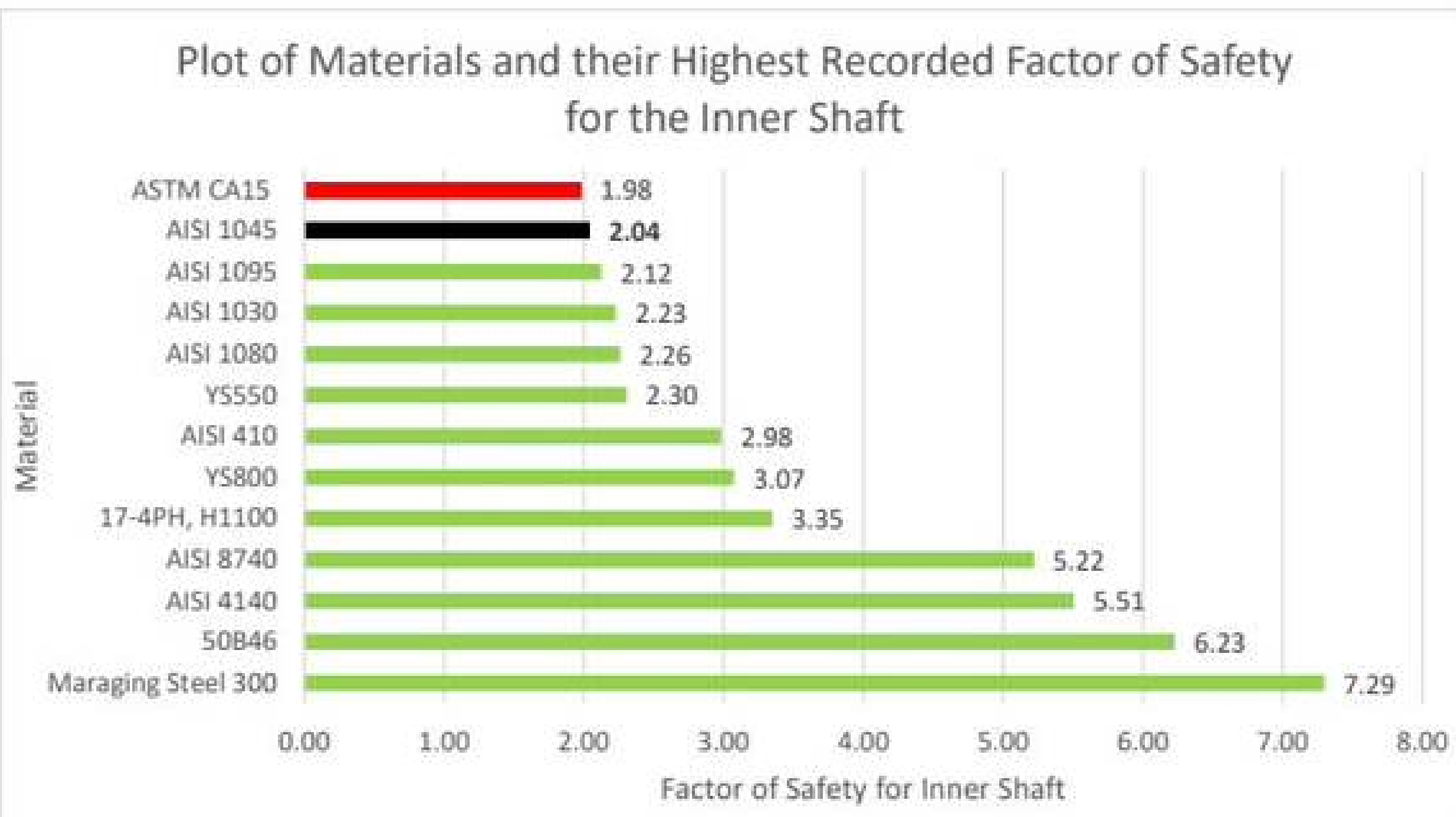

Figure 31: Highest recorded factor of safety for inner shaft across all materials. AISI 1045 is used as the baseline for comparative performance.

Figure 32: Highest recorded factor of safety for outer shafts across all materials. AISI 1045 is used as the baseline for comparative performance.

5.4. Angle of Twist Comparison

The rotational displacement results for each material under applied torque are presented in Table 22 below. By comparing these values to the baseline material, AISI 1045, this section evaluates the torsional stiffness of each shaft design:

Table 22: Angle of twist (θ) results for both shaft types, with percentage differences compared to baseline material AISI 1045.

	Inner Shaft		Outer Shaft	
Material	θ	**Δ% Difference From Baseline Simulations**	θ	**Δ% Difference From Baseline Simulations**
AISI 4140	0.019894400	-3.30	0.001039120	-2.89
AISI 1030	0.019894400	-3.30	0.001039120	-2.89
YS800	0.020198124	-1.83	0.001054899	-1.41
YS550	0.020219933	-1.72	0.001056039	-1.30
AISI 1095	0.020325888	-1.20	0.001061657	-0.78
AISI 1080	0.020325888	-1.20	0.001061657	-0.78
50B46	0.020424318	-0.73	0.001066798	-0.30
AISI 8740	0.020424318	-0.73	0.001066798	-0.30
AISI 1045	**0.020573763**	**0.00**	**0.001070000**	**0.00**
17-4PH, H1100	0.020646200	0.35	0.001078510	0.80
AISI 410	0.020925768	1.71	0.001093070	2.16
ASTM CA15	0.020925800	1.71	0.001093071	2.16
Maraging Steel 300	0.022926196	11.43	0.001197230	11.89

Figures 33 and 34 present the percentage difference in angle of twist for each material compared to AISI 1045. The stiffer materials are highlighted in green and the weaker materials are highlighted in red:

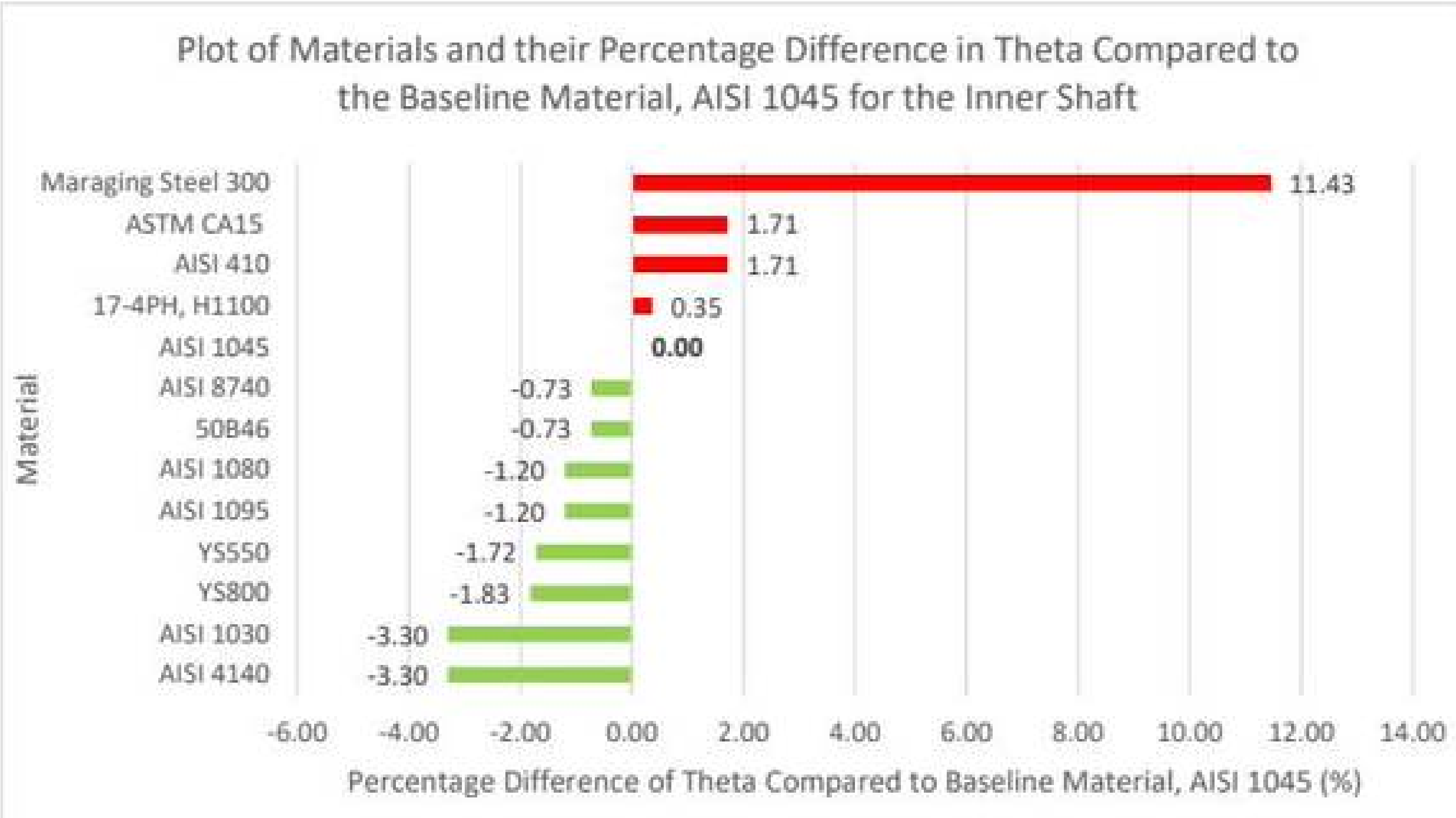

Figure 33: Percentage difference in angle of twist (θ) for inner shaft compared to AISI 1045. Lower values indicate stiffer materials under torsional loading.

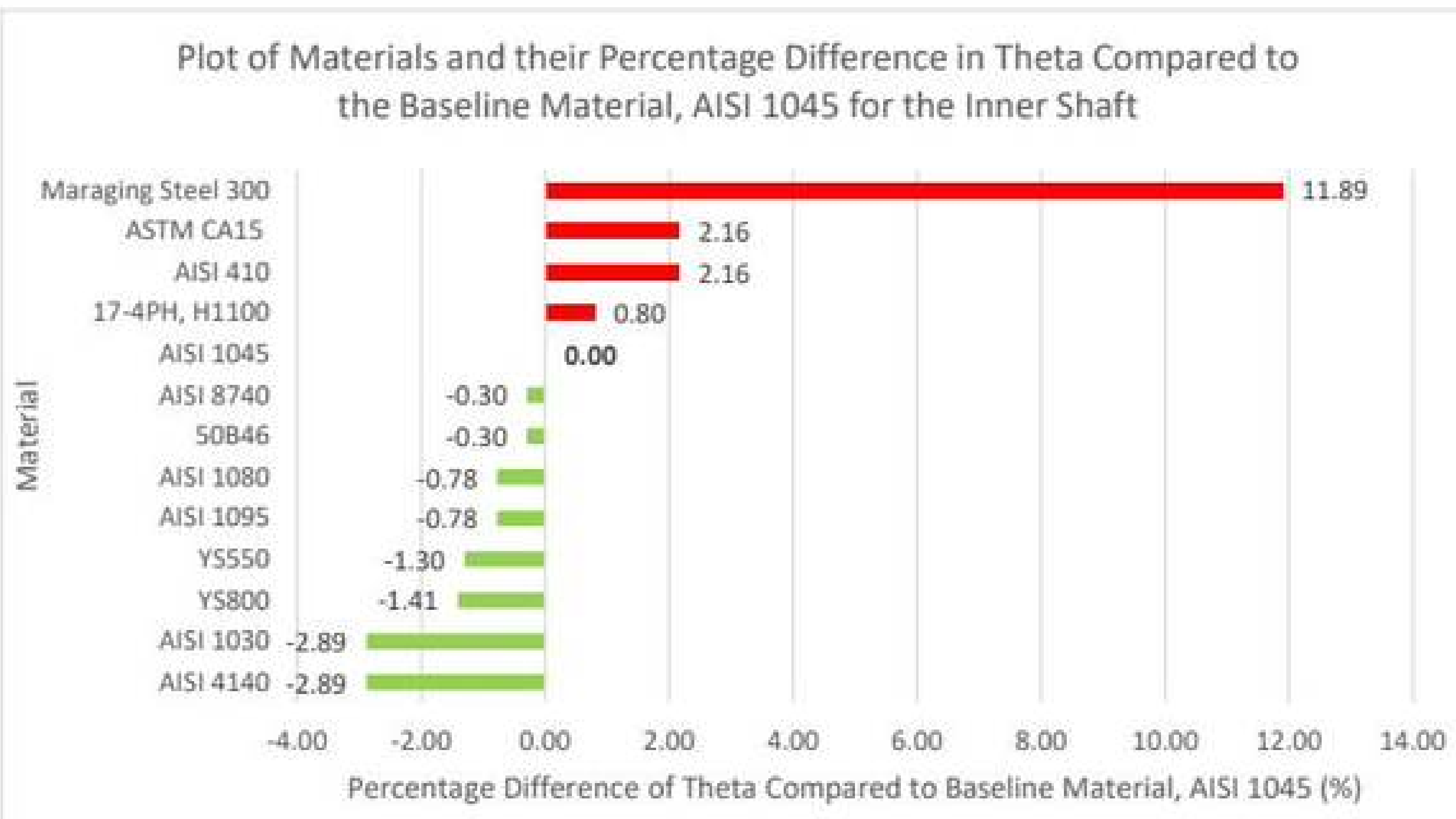

Figure 34: Percentage difference in angle of twist (θ) for outer shaft compared to AISI 1045. Lower values indicate stiffer materials under torsional loading.

6. DISCUSSION

A summary of all parameters analysed for both shafts has been tabulated in Table 23 below:

Table 23: Summary table presenting the percentage change in mass, cost, factor of safety and angle of twist (UR1), for both shafts.

Material	Δ Mass (%)	Δ Cost (%)	FoS (Inner)	FoS (Outer)	$\Delta\theta$ $(\boldsymbol{Inner}, \%)$	$\Delta\theta$ $(\boldsymbol{Outer}, \%)$
AISI 1045	**0.00**	**0.00**	**2.04**	**13.60**	**0.00**	**0.00**
ASTM CA15	-3.06	+64.91	1.98	13.23	+1.71	+2.16
AISI 410	-1.27	+66.88	2.98	19.92	+1.71	+2.16
AISI 1030	0.00	+0.01	2.23	14.88	-3.30	-2.89
AISI 1095	0.00	-1.32	2.12	14.18	-1.20	-0.78
AISI 1080	0.00	+0.01	2.26	15.08	-1.20	-0.78
YS800	0.00	+92.43	3.07	20.51	-1.83	-1.41
YS550	0.00	-1.32	2.30	15.38	-1.72	-1.30
50B46	0.00	+1.35	6.23	41.57	-0.73	-0.30
AISI 4140	0.00	+8.59	5.51	36.76	-3.30	-2.89
AISI 8740	0.00	+17.08	5.22	34.84	-0.73	-0.30
17-4PH, H1100	+0.13	-24.12	3.35	22.37	+0.35	+0.80
Maraging Steel 300	+0.85	+4763.22	7.29	48.66	+11.43	+11.89

6.1. Mass Reduction Potential

The mass comparison study, conducted under fixed shaft geometry, highlighted the potential for weight savings through material substitution alone. As shown in Table 23, only two materials demonstrated a lower total shaft mass compared to the baseline AISI 1045: ASTM CA15, with a 3.06% reduction, and AISI 410, with a 1.27% reduction. These reductions stem from the slightly lower densities of the respective alloys.

However, the remaining materials, including high-strength materials such as 50B46, AISI 4140, and AISI 8740, produced negligible mass reduction due to their density values being comparable to that of AISI 1045. Notably, 17-4PH showed a slight increase in mass at +0.13%, and Maraging Steel 300 had the highest increase at +0.85%, due to its greater density.

These results confirm that substituting material alone, without modifying geometry, yields limited mass reduction. Unlike aluminium or titanium alloys, steels do not vary significantly in density, meaning any effort to reduce weight must also involve geometric optimisation through methods such as wall thinning, hollow sections, or spline optimisation. While this was beyond the scope of the current Project, the findings establish a performance baseline for future lightweighting efforts.

However, the weight savings from ASTM CA15 must be critically weighed against its mechanical drawbacks. Despite being the lightest, ASTM CA15 delivered the lowest FoS of 1.98. Similarly, AISI 410, although slightly lighter, showed only marginal improvement in FoS.

In conclusion, while mass reduction is achievable through selective material choice, it is insufficient. The trade-off between weight and performance must be carefully managed, and lightweighting in DCT input shafts will require a combination of material substitution and geometric optimisation.

6.2. Cost Reduction Potential

Cost is critical in the material selection process, particularly for automotive applications where components are produced at scale. The cost analysis for the inner and outer shafts revealed significant disparities between the candidate materials, with differences exceeding 4700% in extreme cases. The most cost-effective material identified was 17-4PH H1100, giving a 24.12% reduction in total shaft cost relative to AISI 1045. This result is important given that 17-4PH also provided superior performance in both factor of safety and torsional twist, making it a suitable material for low-cost, high-performance substitution.

Other favourable cost-effective solutions were observed for YS550 and AISI 1095, which offered 1.32% cost savings with mechanical performance equal to or slightly better than the baseline. These alloys maintained safety margins while reducing cost, presenting viable alternatives where material price is a priority.

In contrast, several materials presented high-cost penalties. ASTM CA15 and AISI 410, despite marginal or moderate improvements in mass, incurred cost increases of 64.91% and 66.88%, respectively. The most expensive alloy among the tested steels was YS800, with a 92.43% increase. These materials may be acceptable in low-volume or specialised applications, but are unlikely to be justified in high-volume production environments.

While mechanically exceptional, Maraging Steel 300 was deemed economically infeasible for automotive use due to its +4763.22% increase in material cost.

The results present a critical trade-off between mechanical performance and financial feasibility. While high-strength alloys such as 50B46 and AISI 4140 demonstrated exceptional FoS values, they came with cost increases of 1.35% and 8.59%, respectively. These are acceptable in cases where increased safety margin or fatigue resistance justifies the increase in cost, but they must be evaluated within the total system cost.

In summary, cost-effective materials like 17-4PH offer strong mechanical performance at a lower cost than the baseline, while other materials impose financial burdens that are difficult to justify. These findings reinforce the importance of factoring economic constraints into performance-based material selection, especially in high-volume automotive applications.

6.3. Factor of Safety

The factor of safety is an important metric in mechanical design, particularly for rotating shafts subjected to cyclic and torsional loads. It reflects how much stronger a component is than needed for the intended load and is a measure of failure resistance.

FEA simulations under a torque load of 420 Nm revealed that, while von Mises stress levels remained constant across all materials, due to fixed geometry and load conditions, the FoS varied significantly based on each material's yield strength. These variations are critical in determining the suitability of each alloy for use in DCT shafts.

Among all materials tested, Maraging Steel 300 recorded the highest FoS at 7.29 (inner) and 48.66 (outer), due to its high yield strength. However, this performance comes at the expense of cost and practicality, as discussed in Section 6.2.

Excluding Maraging Steel 300, the top-performing alloys in terms of factor of safety were:

- 50B46's FoS = 6.23 (inner), 41.57 (outer)
- AISI 4140's FoS = 5.51 (inner), 36.76 (outer)
- AISI 8740's FoS = 5.22 (inner), 34.84 (outer)

These steels significantly outperformed AISI 1045, which achieved only 2.04 (inner) and 13.60 (outer). These results suggest that high-strength low-alloy steels such as 50B46 and AISI 4140 improve reliability under peak loading and offer potential for downscaling or geometric optimisation without compromising safety.

Materials like 17-4PH, YS800, and AISI 410 demonstrated moderate improvement, with inner shaft FoS values ranging from 2.98 to 3.35, representing a safe mid-range option.

In contrast, ASTM CA15, despite being the lightest material, produced the lowest FoS of 1.98 which was below the design threshold of 2.0 for high-cycle fatigue applications. This performance disqualifies it from consideration in safety-critical roles unless compensatory geometry changes or lower torsional loading conditions are introduced.

Overall, the FoS results confirm that several steel alternatives significantly outperform AISI 1045. These materials offer a good case for substitution in future DCT shaft designs when paired with acceptable cost and stiffness characteristics. High FoS materials also enable future design flexibility, allowing for reduced dimensions or increased torque capacity.

6.4. Angle of Twist

Torsional stiffness, represented in this study by the angle of twist, is a key parameter in transmission shafts. The results were normalised against the baseline material AISI 1045, and percentage differences were used to evaluate relative stiffness.

Most candidate materials demonstrated comparable or improved torsional stiffness over AISI 1045. This outcome is expected given the similar shear moduli across standard carbon and alloy steels. The best-performing materials in terms of twist reduction were:

- AISI 4140 and AISI 1030: −3.30% (inner), −2.89% (outer)
- AISI 8740 and 50B46: −0.73% (inner), −0.30% (outer)

These reductions indicate enhanced torsional rigidity. This stiffness gain is particularly advantageous when paired with high FoS values, as in the case of 50B46 and AISI 4140, reinforcing their suitability as mechanical upgrades over AISI 1045.

Several materials, including YS550, YS800, AISI 1080, and AISI 1095, also demonstrated improved or neutral twist performance. On the other hand, materials such as ASTM CA15 and AISI 410 produced increased angles of twist of +1.71% (inner) and +2.16% (outer). While these differences are not severe in absolute terms, they represent a reduction in stiffness, which may affect dynamic performance in high-speed applications. The increase in is particularly critical when combined with lower FoS, as is the case for ASTM CA15, which ranks poorly on mechanical and functional grounds.

Although exceptional in strength, Maraging Steel 300 recorded the highest increase in twist: +11.43% (inner) and +11.89% (outer). This was a result of its significantly lower shear modulus, which increases the risk of failure. Despite its strength, this material's stiffness characteristics may necessitate overdesign.

Data confirms that most steels tested exhibit similar or improved torsional stiffness relative to AISI 1045. Therefore, torsional stiffness is not a limiting constraint for material substitution in this case.

6.5. Trade-offs Summary

The material evaluation reveals distinct trade-offs between mechanical performance, cost, and mass. Materials such as 50B46 and AISI 4140 emerged as the strongest candidates overall, combining high factors of safety, small difference in mass, and acceptable increases in cost. These alloys offer a balanced route to improving durability and reliability without compromising manufacturability or stiffness.

17-4PH H1100 ranked particularly well from a cost-benefit perspective. Despite only moderate gains in FoS and UR1, its 24.12% cost reduction makes it a practical choice for high-volume applications where affordability is valued. However, its slightly increased mass (+0.13%) and limited strength gains may constrain its use in performance-focused applications.

While ASTM CA15 is the lightest material, it fails to meet the required FoS threshold and displays reduced stiffness, rendering it unsuitable for safety-critical transmission shafts. Maraging Steel 300, though mechanically superior, is economically unjustifiable for mass-market use due to its 4700% cost increase.

Based on all data and simulations, the optimal materials are those that achieve a balance. 50B46 is highly suitable for high-performance applications prioritising FoS, and 17-4PH can be used for cost-sensitive, production-scale contexts.

7. CONCLUSIONS

This study evaluated multiple steel alloys' stress performance and lightweighting potential in DCT input shafts. Alternative materials were assessed through FEA, validated against classical torsion theory, and benchmarked against the baseline AISI 1045 under fixed geometric and loading conditions. The study focused on four key performance metrics: mass, cost, factor of safety, and torsional stiffness.

Results showed that von Mises stress distributions were consistent across all materials due to identical shaft geometry. However, FoS varied significantly as a function of material yield strength. High-strength, low-alloy steels such as 50B46 and AISI 4140 achieved FoS increases exceeding 200% while maintaining acceptable mass and cost trade-offs. Stainless steel 17-4PH demonstrated the greatest cost reduction while marginally outperforming AISI 1045 in all technical parameters.

Mass reduction through material substitution alone proved limited due to the small density ranges between the different steel alloys. ASTM CA15, while offering the greatest weight saving, was rejected due to an FoS below the benchmark. Torsional stiffness remained unchanged across all materials, confirming that stiffness is not a limiting factor in this context.

The study aimed to identify feasible material alternatives to AISI 1045 and demonstrated that strategic material selection can significantly enhance mechanical performance and cost efficiency without requiring geometric modification. By combining validated simulation, multi-parameter comparison, and a materials performance index (MPI), the study provides a robust framework for selecting DCT shaft materials and supports future optimisation in powertrain components.

8. FUTURE WORK

Future work should include fatigue life analysis using S-N or ε-N curve data to evaluate material performance under cyclic loading. Incorporating geometric optimisation, such as hollow shaft designs or spline refinement, could offer further mass reduction. Finally, a sustainability assessment considering embodied energy, CO_2 emissions, and recyclability would support more environmentally responsible material selection.

9. REFERENCES

[1] The International Council on Clean Transportation, *"EU LCV CO_2 Standards 2030,"* Jan. 2019. [Online]. Available: https://theicct.org/sites/default/files/publications/EU-LCV-CO2-2030_ICCTupdate_201901.pdf. [Accessed: Dec. 2, 2024].

[2] J. Dornof, *"CO_2 Emission Standards for New Passenger Cars and Vans in the European Union,"* The International Council on Clean Transportation, May 2023. [Online]. Available: https://theicct.org/wp-content/uploads/2023/05/eu-co2-standards-cars-vans-may23.pdf. [Accessed: Dec. 3, 2024].

[3] European Environment Agency, *"CO_2 Performance of New Passenger Cars in Europe,"* [Online]. Available: https://www.eea.europa.eu/en/analysis/indicators/co2-performance-of-new-passenger. [Accessed: Dec. 12, 2024].

[4] F. Del Pero, M. Delogu, and M. Pierini, *"The effect of lightweighting in automotive LCA perspective: Estimation of mass-induced fuel consumption reduction for gasoline turbocharged vehicles," J. Cleaner Prod.*, vol. 154, pp. 566–577, 2017, doi: 10.1016/j.jclepro.2017.04.013.

[5] Z. You, *"Design of Automotive Mechanical Automatic Transmission System Based on Torsional Vibration Reduction," J. Vibroengineering*, vol. 25, no. 4, pp. 683–697, Jan. 2023, doi: 10.21595/jve.2022.22947.

[6] Q. Chen, J. Wang, and Q. Ahmed, *"Design and Evaluation of a Structural Analysis-Based Fault Detection and Identification Scheme for a Hydraulic Torque Converter," Sensors*, vol. 18, no. 12, p. 4103, 2018. [Online]. Available: https://www.mdpi.com/journal/sensors. [Accessed: Dec. 6, 2024].

[7] D. Stanojević, V. Spasojević, I. Stevanović, and A. Nedić, *"The contemporary automatic gearboxes – Review of the current state and interpretation of advantages and disadvantages of their use with respect to vehicle performance and traffic safety," J. Appl. Eng. Sci.*, vol. 11, no. 2, pp. 89–97, May 2013. [Accessed: Dec. 3, 2024].

[8] Volkswagen AG, *"Technical Service Bulletin: Service Manual for 7-Speed Double-Clutch Transmission 0AM,"* 2012. [Online]. Available: https://static.nhtsa.gov/odi/tsbs/2012/MC-10156073-9999.pdf. [Accessed: Dec. 2, 2024].

[9] C. Andropoulos, *"The PDK Transmission," Excellence Magazine*, Jan. 15, 2015. [Online]. Available: https://www.excellence-mag.com/issues/226/articles/the-pdk-transmission. [Accessed: Dec. 2, 2024].

[10] Volkswagen Media Services, *"Volkswagen DSG – World Standard,"* May 26, 2006. [Online]. Available: https://web.archive.org/web/20060524093743/http://www.volkswagen-media-services.com/medias_publish/ms/content/en/pressemitteilungen/2002/11/22/volkswagen_dsg_-_world.standard.gid-oeffentlichkeit.html. [Accessed: Dec. 2, 2024].

[11] Volkswagen AG, *"The 7-Speed DSG,"* Jan. 21, 2008. [Online]. Available: https://web.archive.org/web/20101216093140/http://www.volkswagenag.com/vwag/vwcorp/info_center/en/themes/2008/01/the_7speed_dsg.html. [Accessed: Dec. 2, 2024].

[12] Nissan, *"Nissan GT-R Specifications,"* Oct. 26, 2006. [Online]. Available: https://web.archive.org/web/20080914214129/http://www.gtrnissan.com/specs.en.us.html. [Accessed: Dec. 2, 2024].

[13] L. Brooke, *"Transmission Technology: The Evolution of Dual-Clutch Systems," SAE International*, Feb. 9, 2008. [Online]. Available: https://web.archive.org/web/20081031232522/http://www.sae.org/automag/technewsletter/080205Tech/04.htm. [Accessed: Dec. 2, 2024].

[14] S. Ye, *"Research on Dual Clutch Transmission (DCT),"* in *Proc. 2022 Int. Conf. Urban Planning and Regional Economy (UPRE)*, vol. 217, Atlantis Press, 2022. [Online]. Available: https://www.atlantis-press.com/proceedings/upre-22/125976957. [Accessed: Dec. 2, 2024].

[15] S. Shrader, *"BMW Is Dropping Dual-Clutch Transmissions for 8-Speed Automatics, Even in M Cars," The Drive*, Jan. 13, 2021. [Online]. Available: https://www.thedrive.com/news/38715/bmw-is-dropping-dual-clutch-transmissions-for-8-speed-automatics-even-in-m-cars. [Accessed: Dec. 2, 2024].

[16] B. T. M. Scheepers, *Automotive Transmission Testing: A New Concept*, DCT Report 2002.018, Eindhoven Univ. of Technology, 2002.

[17] Koenigsegg, *"Technical Specifications – Koenigsegg Regera,"* [Online]. Available: https://www.koenigsegg.com/technical-specifications-regera. [Accessed: Apr. 28, 2025].

[18] I. Barsoum and M. Jonsson, *"Strength Optimization of Induction Hardened Splined Shaft – Material and Geometric Aspects," Int. J. Machine Tools and Manufacture*, vol. 53, no. 1, pp. 94–102, Jan. 2013. [Online]. Available: https://publications.waset.org/6778. [Accessed: Dec. 3, 2024].

[19] I. Dobrevski, *"Dual Clutch Transmission System,"* GrabCAD, Jan. 23, 2016. [Online]. Available: https://grabcad.com/library/dual-clutch-transmission-system-1. [Accessed: Dec. 6, 2024].

[20] F. A. Al-Shammaa and H. F. Kadhim, *"An Analysis of Stress Distribution in a Spline Shaft Subjected to Cyclic Impulsive Load," J. Eng.*, vol. 20, no. 7, pp. 646–657, Jul. 2014. [Online]. Available: https://coeng.uobaghdad.edu.iq/... [Accessed: Dec. 3, 2024].

[21] M. K. Singh, *"Application of Steel in Automotive Industry," Int. J. Emerging Technology and Advanced Engineering*, vol. 6, no. 7, pp. 246–253, Jul. 2016. [Online]. Available: https://www.researchgate.net/publication/313900101. [Accessed: Dec. 2, 2024].

[22] X. Hu and Z. Feng, *Advanced High-Strength Steel—Basics and Applications in the Automotive Industry*, Oak Ridge National Laboratory, ORNL/TM-2021/2047, Apr. 2021. [Online]. Available: https://www.osti.gov/. [Accessed: Dec. 2, 2024].

[23] E. Ghassemieh, *"Materials in Automotive Application, State of the Art and Prospects,"* in *New Trends and Developments in Automotive Industry*, InTechOpen, pp. 365–387. [Online]. Available: https://www.intechopen.com/chapters/36115. [Accessed: Dec. 6, 2024].

[24] F. Kresinsky, E. Leidich, and A. Hasse, *"Different Failure Mechanisms in Keyed Shaft-Hub Connections under Dynamic Torque Load," Procedia Struct. Integrity*, vol. 17, pp. 162–169, 2019. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2452321619301540. [Accessed: Nov. 11, 2024].

[25] K. Sathishkumar and N. Ugesh, *"Finite Element Analysis of a Shaft Subjected to a Load," ARPN J. Eng. Appl. Sci.*, vol. 11, no. 9, pp. 5996–6000, May 2016. [Online]. Available: https://www.arpnjournals.org/jeas/research_papers/rp_2016/jeas_0516_4220.pdf. [Accessed: Dec. 6, 2024].

[26] P. Jayanaidu, M. Hibbatullah, and P. Baskar, *"Analysis of a Drive Shaft for Automobile Applications," IOSR J. Mech. Civil Eng.*, vol. 10, no. 2, pp. 43–46, Dec. 2013. [Online]. Available: https://www.slideshare.net/slideshow/h01024346/43725254. [Accessed: Dec. 6, 2024].

[27] J. H. E. Fox, *An Introduction to Steel Selection: Part 1 – Carbon and Low-Alloy Steels*, Oxford Univ. Press, 1979.

[28] W. C. W., *Stahlschlüssel: Key to Steel*, 15th ed., ASM International, 1989.

[29] UK Steel Association, *Iron and Steel Specifications*, 8th ed., London, UK: UK Steel Association, 1994.

[30] J. Woolman and R. A. Mottram, *The Mechanical and Physical Properties of the British Standard EN Steels (B.S. 970 - 1955)*, Elsevier, 2013.

[31] ASM International, *Guide to Materials Engineering Data and Information*, Metals Park, OH: ASM International, 1986.

[32] ASM International, *ASM Metals Handbook*, 9th ed., Metals Park, OH: ASM International, 1986.

[33] P. Liolios, *"Mohr's Circle in 3 Dimensions,"* Pantelisliolios.com, 2020. [Online]. Available: https://www.pantelisliolios.com/mohr-circle-3d/. [Accessed: Dec. 6, 2024].